

UNIVERSITY OF CALGARY

Exploring Emotion Recognition of Students in Virtual Reality Classrooms Through
Convolutional Neural Networks and Transfer Learning Techniques

by

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A THESIS

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Abstract

In contemporary educational settings, understanding and assessing student engagement through non-verbal cues, especially facial expressions, is pivotal. Such cues have long informed educators about students' cognitive and emotional states, assisting them in tailoring their teaching methods. However, the rise of online learning platforms and advanced technologies like Virtual Reality (VR) challenge the conventional modes of gauging student engagement, especially when certain facial features become obscured or are entirely absent. This paper explores the potential of Convolutional Neural Networks (CNNs), specifically a customized training approach model adapted from the ResNet50 architecture, in recognizing and distinguishing subtle facial expressions in real-time, such as neutrality, boredom, happiness, and confusion. The novelty of our approach is twofold: First, we optimize the power of CNNs to analyze facial expressions in digital learning platforms. Second, we innovate for the context of VR by focusing on the lower half of the face to tackle occlusion challenges posed by wearing VR headsets. Through comprehensive experimentation, we compare our model's performance with the default Residual Neural Network 50 (ResNet50) and evaluate it against full-face and VR-occluded face datasets. Ultimately, our endeavor aims to provide educators with a sophisticated tool for real-time evaluation of student engagement in technologically advanced learning environments, subsequently enriching the teaching and learning experience.

Keywords: *emotion recognition, facial expression, virtual reality, virtual learning environment, machine learning, deep learning, CNN, transfer learning, ResNet50, VGG19, MobileNet.*

Preface

The journey of this thesis started with a profound observation and questions in the changing landscape of education; as we enter the digital age, understanding and preserving the nuances of human engagement are becoming increasingly crucial especially in learning environments. This work was driven by the realization that, while VR classrooms offer groundbreaking opportunities for immersive learning, they also bring about new challenges in capturing and interpreting the subtle cues of student engagement – cues that play a critical role in the educational experience.

Throughout my research, I had the privilege to explore the intersection of technology, psychology, and education. I delved into how artificial intelligence (AI) can bridge the gap between traditional classroom settings that provide rich feedback and the dynamic, often non-verbal, interactions within VR spaces. This thesis goes beyond academia, it ventures into a future where technology enhances human connections rather than diminishing them.

The development of the CNN model presented in this thesis is a result of integrating innovation with pedagogical empathy; specifically, it leverages the ResNet50 architecture. It shows a commitment to honor the silent language of student's expressions such as, boredom, neutrality, confusion, and happiness, and converting it into valuable data that can empower educators.

The following chapters delve into the process of collecting data, experimentation and meticulous creating a model that goes beyond observation to deep understanding. This thesis represents both my academic journey and a vision for the future of education.

Acknowledgement

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To the members of our research team: your innovative ideas and contributions during our weekly meetings have not only pushed the boundaries of our field but also sparked a wellspring of inspiration. It is my sincere hope that this work will significantly contribute to the ongoing discussions about the incorporation of virtual reality in education and serve as a catalyst for further exploration into the boundless possibilities that emerge when technology is harmonized with human insight.

I also wish to acknowledge all educators and students who stand at the vanguard of learning, adeptly adapting to and shaping the digital revolution in education. May the findings within these pages provide robust support and enrich the educational experiences in the virtual worlds we are collectively bringing to life.

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List of Symbols, Abbreviations and Nomenclature

Abbreviations	Definitions
VR	Virtual Reality
CNNs	Convolutional Neural Networks
ResNet	Residual Neural Network
AI	Artificial Intelligence
ML	Machine Learning
VGG	Visual Geometry Group
RGB	Red, Breen and Blue
RGB-D	Red, Breen and Blue-Depth
FER+	Facial Emotion Recognition Plus
RAF-DB	Real-world Affective Faces Database
DL	Deep Learning
KDEF	Karolinska Directed Emotional Faces
MMI	Macro- and Micro-Expressions
CK+	Cohn-Kanade
MC-loss	Mutual-Channel Loss
IoT	Internet of Things
AR	Augmented Reality
PCA	Principal Component Analysis
RL	Reinforcement learning
GPUs	Graphics Processing Units
TPUs	Tensor Processing Units
RNN	Recurrent Neural Networks
LSTMs	Long Short-Term Memory Networks
VAEs	Variational Autoencoders
GANs	Generative Adversarial Networks

NLP	Natural Language Processing
BERT	Bidirectional Encoder Representations from Transformers.
GPT	Generative Pre-Trained Transformer
CapsNets	Capsule Networks
ILSVRC	Large Scale Visual Recognition Challenge
ReLU	Rectified Linear Unit
FC	Fully Connected
API	Application Programming Interface
ONEIROs	Open ended Neuro Electronic Intelligent Robot Operating System
OpenCV	Open-Source Computer Vision
REB	Research Ethics Board
PNTA	Prefer Not To Answer
Email	Electronic Mail
FPS	Frames Per Second
SoftMax	Soft argmax
val_loss	Validation Loss
LR	Learning Rate
Min	Minimum
HDF5	Hierarchical Data Format version 5
Wandb	Weights and Biases
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
Eq.	Equation
F	Full Face
H	Half Face

Chapter 1: Introduction

1.1 Introduction

Over the years, there have been shifts from teaching methods using chalkboards to more technologically advanced learning environments. Amidst these changes, understanding student engagement remains a high priority for teachers. Research indicates that there is a correlation between student engagement and academic performance (An et al., 2023).

In classrooms where both students and teachers are physically present, teachers have had various ways to assess the student's engagement. One of the traditional methods has been through observing non-verbal cues, especially facial expressions. Teachers have long relied on these cues as indicators of student's cognitive and emotional states (Ekman & Friesen, 1971). Students' faces give teachers a lot of information which helps them adjust their teaching methods depending on how students react to the material being taught. However, with the rise of online education, this direct observation is often limited or non-existent, especially when students choose to keep their cameras off or face connectivity issues (Sitzmann et al., 2006).

Keeping track of student's reactions, emotions and engagements on online learning platforms can be quite challenging. It deprives teachers of the real time feedback mechanism that allows them to adapt their teaching strategies on the spot. Potentially creates a disconnect between instruction and student's needs. There have been some efforts to tackle this problem by utilizing AI and machine learning (ML) algorithms to analyze how engaged students are in classrooms (Calvo et al., 2015). However, these technologies are still in their stages, as they cannot fully replicate the depth of human understanding that comes from direct observation.

While it is well documented that online learning environments have limitations, especially when it comes to supporting nuanced verbal cues, it is important to recognize that virtual classrooms cause more than just negative effects on education. Indeed, many researchers have emphasized the benefits and potential impact of learning experiences (Zheng et al., 2015). For example, the flexibility and accessibility provided by platforms can often outweigh the advantages of classrooms, particularly in situations where geographic or physical limitations pose challenges to conventional learning (Mohamed, 2004).

Two expressions often encountered by teachers, "Boredom" and "Neutrality," can be challenging to distinguish but are crucial for effective teaching (Kunter et al., 2008). When a student has a neutral expression on their face, it may indicate that they are calmly attentive. They could be quietly absorbing information, reflecting on its meaning, or mentally organizing it in a way that makes sense to them. In cases, a neutral facial expression can indicate an educational experience where the student is effectively engaging with the material at hand (Picard et al., 2004). On the other hand, if a student appears bored, it can raise concerns for teachers about disengagement or lack of interest in the topic being discussed. Unlike a neutral expression, boredom often requires adjustments in teaching methods, such as introducing interactive elements or even changing the direction of the lesson plan altogether (Larson & Richards, 1991). Distinguishing between these two expressions may be crucial for teachers striving for effective teaching (Kunter et al., 2008). Although there are many opportunities for virtual learning, it is important to be aware of the difficulties because these opportunities include the absence of nonverbal cues like facial expressions, emotions, reactions, and active participation.

Our work focuses on the limitation posed by missing facial cues in virtual education settings. We employ CNNs, leveraging the ResNet50 architecture (He et al., 2016), to accurately analyze facial expressions such as neutrality and boredom without starting from scratch.

In the era of technological advancements, VR can bring about changes in various aspects of our lives. Both educational technologists and pedagogical theorists are currently exploring how VR can create captivating and effective learning environments. Over time, VR has transformed from an entertainment and gaming concept into a tool with wide ranging implications across industries such as healthcare, manufacturing, and notably education (Pei et al., 2019). With VR, learners can now engage with 3D environments where abstract concepts can be brought to life. Imagine a biology student walking through the system of the body, or a history student freely wandering around an ancient civilization while interacting with its culture. This feeling of being present in these worlds could potentially revolutionize how we engage in education by making it an experiential and successful process. While researchers have studied the impacts of VR in fields like engineering, gaming and medicine, its effectiveness in educational settings is still ongoing research.

As we envision the future, we expand our research to explore the integration of VR in educational settings, such as a virtual classroom where all students join through wearing a VR headset. VR integration in education is tantalizing and growing in popularity because it provides immersive teaching and learning experiences. However, it also presents challenges such as barriers, potential health effects and difficulties with analyzing expressions due to obstructions caused by VR headsets (Radianti et al., 2020). Therefore, this paper does not only focus on highlighting the benefits, but also addresses the obstacle of occlusion when analyzing users' facial expressions (Figure 1-1). To address this issue, our CNN model has been cleverly designed

to focus on the visible half of the face— the lower half. This adaptation ensures that our custom approach model remains practical and relevant for analyzing facial expressions in these advanced immersive environments.

Our research covers the model's architecture, development, experiments, and results. We aim to offer educators a tool for real-time facial expression analysis during lectures, providing reports without storing student images, ensuring privacy (De la, 2022).



Figure 1-1: A person wearing a VR headset (Meta Quest Pro), showing occlusion in the upper half of the face. This is generally the case for any currently available VR headsets in the market.

1.2 Motivation

This research aims to address a gap in the field of education, specifically in virtual reality classrooms. As technology continues to advance, new opportunities and challenges arise in the realm of education. The transition from traditional chalk and talk teaching methods to digital platforms has brought innovative ways to engage students but has also made some traditional assessment techniques less effective or outdated. Our motivation is centered around recognizing and addressing this dilemma. A key aspect of teaching is the ability to understand and respond to student engagement. Educators have honed this skill over generations by interpreting verbal cues, particularly facial expressions. These subtle cues provide insight into a student's emotional

state, helping educators know when adjustments to their teaching approach are needed. However virtual reality classrooms pose a challenge as they can obstruct or completely hide expressions due to VR headsets. This raises the question; how can educators assess student engagement when visual cues that were traditionally relied upon are not accessible?

Our work is motivated by the desire to bridge the gap between verbal communication and the immersive nature of VR in education. We are driven by the challenge of creating a solution that can adapt to VR headsets limitations while providing educators with real time analysis of student engagement through their facial expressions. Additionally, accurately distinguishing between expressions of boredom and neutrality which are crucial for assessing engagement levels adds another layer of complexity. Our research aims to identify these expressions, empowering educators to make decisions that enhance the learning experience.

The main motivation behind our research extends beyond achievement – it focuses on its value. By improving the ability to monitor engagement, we strive to contribute to the advancement of education as a captivating, effective and experiential mode of learning. Through our works lens, we envision a future where the digital divide is reduced, and teachers gain nuanced insights into student engagement, ensuring that no students emotional or cognitive state goes unnoticed in VR based environments.

1.3 Future Applications of the Custom Approach Model in VR Classrooms

The advent of VR in educational settings has opened new vistas for immersive and interactive learning experiences. The integration of emotion recognition technologies, particularly the custom approach CNN model developed in this thesis, holds significant potential in revolutionizing these virtual learning spaces. This section envisions a scenario where our

custom approach model is seamlessly integrated into a VR classroom, enhancing the teaching-learning dynamic through real-time emotional feedback or post-class facial expression analysis.

Scenario:

Imagine a VR classroom where each student, equipped with a VR headset, is immersed in a 3D educational environment. The class is diverse, with students from various backgrounds engaging in a lesson designed to accommodate different learning styles. Integrated within each VR headset is a compact camera system, discreetly positioned to capture the students' facial expressions without causing any discomfort or distraction. As the lesson progresses, the custom training approach CNN model embedded within the classroom's educational software analyzes the facial expressions of each student in real-time or captures their facial expressions for post-class analysis. By focusing on the lower half of the face, the model overcomes the occlusion challenge posed by the VR headsets. It identifies subtle expressions indicative of emotions like confusion, boredom, neutrality, and happiness.

If it is in real-time, the model's output is relayed to the teacher's VR interface, where a dashboard displays an aggregated emotional overview of the class. This immediate feedback allows the teacher to adjust the pace, content, and delivery of the lesson dynamically. For instance, if a significant number of students show signs of confusion or boredom, the teacher can introduce interactive elements, revisit complex topics, or initiate a Q&A session. If it is post-class, the system captures students' facial expressions for subsequent analysis, which the teacher can access after the class. The resulting reports provide a comprehensive view of each student's emotional responses in chart format. This data then enables the teacher to pinpoint moments during the class when a notable increase in boredom or disengagement occurred.

This integration of emotion recognition technology cultivates an empathetic educational environment. Students, aware that their emotional feedback is valued and acted upon, feel more connected and engaged. Teachers, equipped with insights into their students' emotional states, can foster a more inclusive and responsive learning atmosphere. The envisioned application of the custom approach CNN model in VR classrooms paves the way for a new era in digital education. This technology, with its potential to understand and respond to students' emotional states, can significantly enhance the effectiveness of VR-based learning. It promises a future where education is not only immersive and interactive but also emotionally intelligent and empathetic.

1.4 Research Questions

Exploring and innovating the field of education through emotion recognition in VR classrooms offers exciting opportunities. The combination of technology and teaching in VR raises questions about the effectiveness, design and ethical considerations of automated systems that aim to understand and respond to student engagement through facial expressions. This research focuses on using customized training approach CNN to investigate the capabilities and implications of emotion recognition, within a context. We will examine the challenges involved and explore how it can impact the teaching and learning experience. By addressing these questions, we aim to develop an emotion recognition model that is both technically, robust, and pedagogically valuable, while also being ethically responsible. These research questions will guide our data collection process, methodology, experimental result, and ultimately shape our conclusions regarding integrating emotion recognition technology into VR classrooms.

1.4.1 Primary Research Question:

- a) How does the performance of the default ResNet50 architecture compare with other prominent CNN models like Visual Geometry Group 19 (VGG19) and MobileNet when evaluated on the specific dataset of facial recognition in VR classroom settings?
- b) How effectively can a customized training approach CNN adapted from the ResNet50 architecture recognize and differentiate between subtle facial expressions, such as neutrality, boredom, happiness, and confusion, in a VR classroom setting?

1.4.2 Subsidiary Questions:

- a) What are the limitations of current emotion recognition technologies when applied to VR environments, and how does our customized training approach model address these challenges?
- b) How does occlusion caused by VR headsets affect the accuracy of facial expression recognition, and what techniques can improve recognition performance for the visible portions of the face?
- c) What is the comparative performance of the default ResNet50 architecture versus the customized training approach model in recognizing facial expressions from a full facial dataset and a dataset partially occluded by VR headsets?
- d) How can data augmentation techniques enhance the robustness of the facial recognition model when dealing with the diversity of facial expressions in VR classroom scenarios?

1.4.3 Technical Research Question:

- a) What are the optimal preprocessing techniques required for training a CNN on grayscale images as compared to Red, Green and Blue (RGB) images for the purpose of emotion recognition?

1.4.4 Implementation and Impact Question:

- a) What are the practical implications of integrating an emotion recognition system into VR educational software, and how might it alter the landscape of digital learning?

Chapter 2: Related Literature

2.1 Background

Traditional face to face classrooms has benefited from the nuanced interaction of signals and expressions between teachers and students. Research has demonstrated that emotions like boredom, happiness and confusion can significantly influence the learning process. This non-verbal cue serves as a means of communication, enabling teachers to make real time adjustments to their course materials. For instance, when a teacher notices confusion or disengagement among students through their expressions, they may take it as a sign to pause, revisit complex concepts, directly ask if further clarification is needed or provide examples. Research in psychology has shown that these teaching methods are beneficial because they emphasize the significance of involvement and prompt feedback for instruction (Hattie & Timperley, 2007). These non-verbal cues have played an invaluable role in assessing student progress in real time. They provide an immediate feedback system that can assist teachers in identifying the range of student engagement levels, from those who are fully engaged with the content, to those who are uninterested or struggling. Disregarding these cues could lead to a learning environment where students either mentally disengage or go through heightened levels of anxiety and frustration, which could potentially affect students' grades at the end of a session.

2.2 VR in Online Learning

Teachers have reported that integrating VR into learning environments captures students' interest, boosts their creativity, motivation, and technological literacy, while also aiding their comprehension of concepts. Similarly, students have displayed attitudes towards learning through VR platforms like Google Classroom. Moreover, a virtual reality learning environment

was found to be promising for developing classroom management skills and enhancing teacher resilience. Both pre-service teachers and school based teacher educators acknowledge the value of VR in teacher training curricula (Cooper et al., 2019).

By leveraging VR as a tool, there is an opportunity to enhance student engagement, motivation and comprehension of challenging subject matter. (Yildirim et al., 2020) delve into the perspectives of pre-service teachers regarding the incorporation of VR as a resource. To accomplish their goal, they gather data through surveys and interviews with service teachers to capture their perspectives and experiences with VR in classrooms.

Recently, VR has been used as a tool in games - serious games, with the goal of improving education. (Zhao et al., 2020) focused on developing a VR game specifically designed for manufacturing education. They aim to explore the potential of VR technology in improving manufacturing education by creating an interactive learning experience. Furthermore, they evaluate its effectiveness based on user feedback and assessment of learning outcomes. The paper shed light on how the VR game affects students learning experience and knowledge retention, in manufacturing education.

VR-based classrooms also impact student engagement and motivation, resulting in increased interest and active participation in science lessons. (Kasapakis et al., 2023) delved into the benefits of incorporating VR technology into education specifically focusing on the impact of realistic nonverbal cues on the overall learning experience. As a result, they recommended that educators and designers consider incorporating these cues into VR based educational experiences to optimize learning outcomes. Similarly, (R. Liu et al., 2022) focused on investigating the impact of a VR based classroom on students' performance in science lessons. To conduct their experiment, they divided students into two groups: one receiving classroom instruction, while the

other experiencing instruction within a VR based classroom. The outcome of their work shows that students in a virtual classroom perform better than those who received instructions in physical education class.

2.2.1 VR Occlusion Facial Recognition

(Houshmand & Mefraz Khan, 2020) propose a geometric model to simulate occlusion resulting from a Samsung Gear VR headset and use transfer learning with VGG and ResNet pretrained networks to recognize facial expressions in the presence of severe occlusion. They fine-tuned the networks on Facial Emotion Recognition Plus (FER+) and Real-world Affective Faces Database (RAF-DB) datasets to improve the performance of their model. Their proposed approach achieved comparable results to the existing methods while training on modified benchmark datasets that simulate realistic occlusion resulting from wearing a VR headset.

In a paper by (Georgescu & Ionescu, 2019), the authors discuss a method for recognizing facial expressions of occluded faces using CNNs. Their method involves training CNNs on modified training images where the upper half of the face is completely occluded. Their models were fine-tuned in two stages, first on images with full faces and then on images with the upper half of the face occluded. The experimental results demonstrate the effectiveness of the proposed method in recognizing facial expressions of occluded faces, improving the accuracy of facial expression recognition in challenging conditions. The models trained on lower-half faces have higher accuracy rates than models trained on entire faces. In the experimental result, they claimed that the lower part of the face provides enough clues to accurately predict facial expressions. Their result shows that models' predictions on lower-half faces are up to 13% higher than baseline models trained on entire faces, while models' predictions on lower-half faces are no more than 10% under baseline models' predictions on full faces.

2.3 CNN Models for Facial Recognitions

The advent of DL, particularly CNNs, has significantly advanced facial recognition technology. This progress is crucial in VR environments where facial recognition often faces the challenge of occluded or partially visible faces. The introduction of CNNs has been a game-changer in this aspect due to their ability to extract and learn hierarchical feature representations from visual data (Lecun et al., 1998).

2.3.1 ResNet50

The integration of facial recognition systems into institutions has demonstrated numerous advantages. The ResNet50 architecture, a neural network has proven to be highly effective in recognizing images. When it comes to recognition, researchers have successfully used transfer learning with deep learning models by fine-tuning it on specific educational datasets. (Alhanaee et al., 2021) explored the use of different CNN models with transfer learning for facial recognition smart attendance system.

The effectiveness of ResNet50 in facial recognition was explored in a study by (Pandya H et al., 2019), highlighting its robustness in various challenging environments. The researchers utilized ResNet50 to train on a dataset consisting of thousands of images captured under lighting conditions, angles, and expressions. By comparing it with other CNN architectures, they found that ResNet50 demonstrated performance in terms of both speed and accuracy, making it an excellent choice for their facial recognition tasks. Similarly, (Benedicta Nana Esi et al., 2022) carried out a comprehensive analysis by comparing ResNet50, AlexNet and Inception-V3 architectures on masked face recognition. Their work highlighted ResNet50's superiority over other deep learning architectures in terms of both accuracy and computational efficiency.

However, while the potential of ResNet50 in facial recognition is evident, challenges persist. (Hwang et al., 2023) study shed light on potential vulnerabilities in ResNet50-based facial recognition systems, emphasizing the need for adversarial training to bolster security.

2.3.2 VGG19

VGG19 model, known for its depth and the use of small convolutional filters, has made substantial contributions to the field. Developed by (Simonyan & Zisserman, 2015), VGG19's architecture allows for capturing finer details, which is crucial in facial recognition tasks. Its effectiveness in feature extraction from facial images has been validated in numerous studies, including work by (Khajuria et al., 2023), who demonstrated its proficiency in recognizing facial emotions, a key aspect of understanding student engagement in virtual classrooms.

(Dong et al., 2023) proposes an enhanced algorithm for facial expression recognition using a CNN. The algorithm utilizes the VGG network structure and incorporates a Soft pool technique to improve the extraction of facial expression features. An attention mechanism is also integrated into the network to enhance its ability to recognize facial expressions. The algorithm achieves a recognition rate of 86.1% on the RAF-DB dataset, surpassing the performance of the original VGG network on the same dataset. The research suggests that the modified network is effective for facial expression recognition.

In a paper by (Abdallah et al., 2021), the authors explore the use of deep transfer learning with the VGG-16 model for recognizing student behavior in the classroom setting. They used VGG model as the base model for feature extraction, and transfer learning is applied to fine-tune the model for behavior recognition. The study aims to improve the accuracy of student behavior recognition in the classroom using deep transfer learning techniques. Their experimental results

show that their proposed method using deep transfer learning with VGG-16 achieves promising performance in recognizing student behavior in the classroom.

2.3.3 MobileNet

(Nan et al., 2022) proposed a lightweight A-MobileNet model. First, they introduced an attention module into the MobileNetV1 model to enhance the local feature extraction of facial expressions. Then, the center loss and SoftMax loss are combined to optimize the model parameters to reduce intra-class distance and increase inter-class distance. They compared their custom model with the original MobileNet series models; their method significantly improves recognition accuracy without increasing the number of model parameters. Compared with others, A-MobileNet model achieves better results on the FER+ and RAF-DB datasets.

(Han et al., 2022) in their work deals with the existing drawbacks of facial expression recognition methods that is difficult for network learning on cross-dataset facial expressions, etc., on how they did not consider the inter-class and intra-class features in image classification. They raise a symmetric mode to extract the inter-class features and intra-class diversity features, and then propose a triple-structure network model based upon MobileNet V1, which they trained via a new multi-branch loss function. Their Experimental result on the Karolinska Directed Emotional Faces (KDEF), Macro- and Micro-Expressions (MMI), and Cohn-Kanade (CK+) datasets show that the accuracy of facial expression recognition using the proposed model is 1.224%, 13.051%, and 3.085% higher than that using Mutual-Channel Loss (MC-loss) VGG16, respectively. They proved that their raised triple-structure network model reaches better performance than dozens of state-of-the-art methods.

Chapter 3: Introductory Concepts

3.1 Machine Learning Overview

ML is a branch of AI that focuses on creating algorithms of learning from data and making predictions or decisions (Shaveta, 2023). It allows computers to handle new situations by analyzing, self-training, observing, and gaining experience. ML utilizes data patterns to improve performance in tasks with minimal human intervention. The process of ML involves gathering data, which then undergoes thorough preprocessing to ensure its cleanliness and usefulness (ÇeliK, 2018). Once the dataset is prepared, a model that aligns its capabilities with the problem at hand would be selected and trained by adjusting its parameters to enhance its performance.

3.1.1 Machine Learning Applications

The widespread application of machine learning has significantly transformed different sectors by changing how they operate. In healthcare, algorithms are pioneering advancements in predictive diagnostics and personalized medicine (Woodman & Mangoni, 2023). The finance industry uses these systems to handle tasks such as credit scoring and fraud detection. They analyze amounts of data to ensure the security of transactions and investments. Retail giants utilize machine learning to personalize customer experiences and improve operations, while the transportation sector benefits from optimized routes and the emergence of self-driving vehicles. In the manufacturing industry, machine learning plays a major role in maintenance and streamlined supply chains, leading to improved efficiency. The entertainment industry has also undergone a transformation with ML systems offering personalized content that matches users' preferences. Customer service has been revolutionized by chatbots and virtual assistants that provide support enhancing customer interaction (V. & S., 2023).

One transformative application of machine learning is in computer vision (Khan & Al-Habsi, 2020), where machines can interpret and understand information. Facial detection and recognition systems are now widely used in settings like security and surveillance, user authentication and photo tagging. These advanced machine learning models can accurately detect, analyze, and recognize faces from images or video feeds even when faced with challenges such as angles, lighting conditions or obstructions. The potential of this capability is truly remarkable as it expands the limits of what can be achieved in security systems, targeted advertising, user interface design, interactive gaming and VR (Chen, 2023; Losada Arias et al., 2022). It enables the creation of intuitive experiences that are tailored to users' facial expressions and reactions.

3.1.2 Challenges in Machine Learning

The pursuit of achieving excellence in ML is confronted by a variety of challenges, some arising from the nature of data and algorithms, while others originating from ethical and practical considerations (Jordan, 2018). Ensuring the quality and quantity of data remains a challenge, as the effectiveness of ML algorithms is directly linked to the quality of the data they are provided. Incomplete or biased datasets can result in models that are either overfit or underfit, limiting their usefulness in real world scenarios. Furthermore, the opacity associated with algorithms, especially deep learning ones, raises concerns about transparency and makes it difficult to understand and trust their decision-making processes. Privacy issues and ethical dilemmas further complicate matters when using personal data for training models, requiring navigation through consent and data protection regulations (Ge, 2023). The evolving landscape of cybersecurity also poses a risk as machine learning systems become both targets for sophisticated cyber-attacks and tools used to carry them out.

3.1.3 The Future of Machine Learning

The future holds promise for ML, envisioning a world where its integration seamlessly becomes a part of our everyday lives. The trajectory of ML points towards algorithms that possess self-improvement capabilities, pushing the boundaries of what machines can learn and achieve. As AI and quantum computing intersect, we can expect to witness advancements in processing power and algorithmic efficiency that will open possibilities. The combination of ML with emerging technologies, like the Internet of Things (IoT) (Szydlo et al., 2020) and augmented reality (AR) (Cho et al., 2022) points towards a future where intelligent environments and enhanced realities become commonplace. The rise of AI is also expected as there is a growing need to develop algorithms that are fair, accountable, and transparent, not just from a technical standpoint but as a societal requirement. Lastly, in the face of challenges such as climate change and pandemics (Sudakov & Titov, 2023), machine learning could play a role in creating innovative solutions that are both necessary and sustainable potentially allowing us to forecast events and drive the revolution towards sustainability.

3.1.4 Types of Machine Learning

The field of ML is diverse and multifaceted, with various approaches tailored to solve distinct types of problems. These approaches can be classified into three principal categories, each with its methodologies, applications, and theoretical underpinnings.

a) Supervised Learning

The core of supervised learning lies in the concept of approximating functions. The main objective is to deduce a function from labeled training data, which comprises a collection of training examples. Each example consists of an input vector, and its corresponding

desired output value referred to as a label or target. Supervised learning algorithms analyze this training data and generate an inferred function that can be used for mapping examples. This framework is widely used in machine learning and includes algorithms like linear regression, logistic regression, support vector machines, neural networks, and random forests.

The effectiveness of supervised learning depends on the quality and quantity of labeled data. It has been successfully applied to problems such as image recognition, speech recognition, medical diagnosis, stock market analysis, and more. Training in supervised learning often requires computational resources, especially for complex tasks, like deep learning—where large datasets and extensive neural networks are common practices (Sah, 2020).

b) Unsupervised Learning

Unsupervised learning, in contrast to supervised learning works with data that does not have predefined labels. The algorithms aim to uncover structures within the data, such as grouping or clustering of data points. Techniques like k-means clustering, hierarchical clustering and Principal Component Analysis (PCA) fall under unsupervised learning. These methods excel at identifying patterns and correlations in the data that may not be immediately evident (Gentleman & Carey, 2008).

Unsupervised learning is particularly valuable for data analysis, reducing dimensionality and extracting features. For example, market researchers utilize clustering to identify segments within their customer base, while biologists may employ it to categorize plant or animal species based on different attributes.

c) Reinforcement Learning

Reinforcement learning (RL) is a field of ML inspired by behavioral psychology. In this approach a model learns to make decisions by taking actions in an environment with the goal of maximizing cumulative rewards. The learner is not told which actions to take but instead must discover which actions yield the most reward by trying them. This type of learning involves a feedback mechanism based on rewards commonly referred to as the reward signal (Ogunfowora & Najjaran, 2023).

RL finds its applications in fields such as robotics, where robots acquire skills through trial and error or in gaming (Saadat & Zhao, 2022), where AI agents learn and excel at games like miniXCOM or chess. It is also utilized in resource management, navigation and making real time decisions in complex ecosystems.

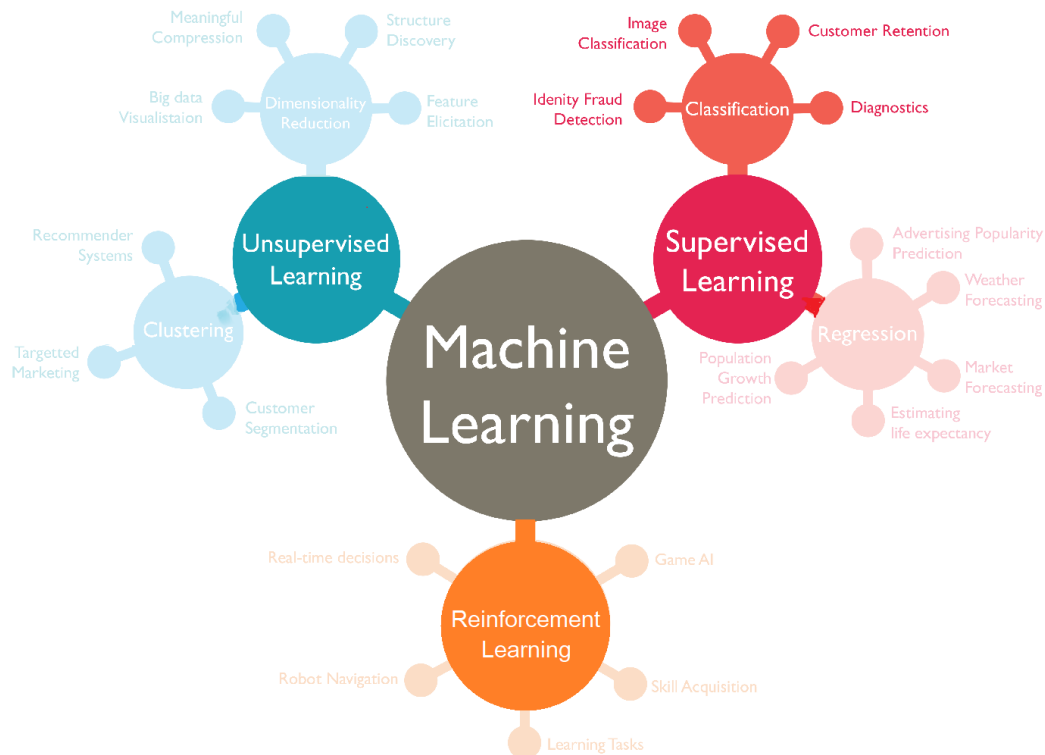


Figure 3-1: Machine Learning Branches. Source:(Towardsdatascience.com.)

3.2 Deep Learning Overview

Deep Learning has revolutionized the field of AI by pushing the boundaries of what can be achieved with machine learning (Buntine, 2020). Essentially, DL uses neural networks that are deeper, and more complex compared to traditional ones. These deep networks consist of layers of interconnected nodes or neurons, enabling them to analyze and learn from vast amounts of data in a way that resembles human cognitive processes. The true power of learning lies in its ability to grasp multiple levels of representation and abstraction, allowing it to make sense of diverse types of data such as images, sound, and text. Unlike simple machine learning algorithms that may struggle with analyzing raw data, deep learning models can progressively learn through their internal layers of concepts, starting from detecting edges in visual data, to recognizing shapes, then to complex objects, and finally identifying scenes and contexts.

Although deep learning models have achieved accomplishments, they also have their challenges. They rely on amounts of labeled data for effective training and demand substantial computational resources. It is often necessary to use Graphics Processing Units (GPUs) or Tensor Processing Units (TPUs) for training within a reasonable timeframe (Y. E. Wang et al., 2019). Furthermore, the complexity of these models can result in a lack of transparency regarding their decision-making process, referred to as the "black box" dilemma.

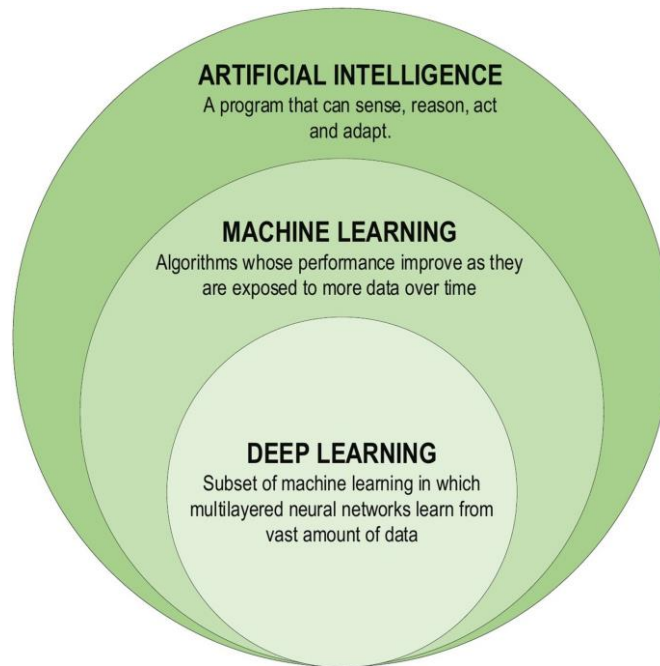


Figure 3-2: Deep Learning Family. Source: (Alzubaidi et al., 2021)

3.2.1 Applications and Future Directions

Deep learning has had effects in different fields. In tasks like image and speech recognition, deep neural networks have sometimes surpassed human performance. In natural language processing, DL has facilitated advancements in machine translation, sentiment analysis, conversational agents, poem and poetry generation (Elzohbi & Zhao, 2023).

Looking ahead, the field of DL is progressing towards architectures that require less data and computational power. There is a growing emphasis on models that offer greater interpretability and techniques that can learn more efficiently through unsupervised or semi supervised methods. Emerging research areas such as few-shot learning (Ye et al., 2022) (where models learn from data) and transfer learning (where knowledge from one task enhances learning in another) hold potential to expand the scope and capabilities of deep learning (Talaie Khoei et al., 2023).

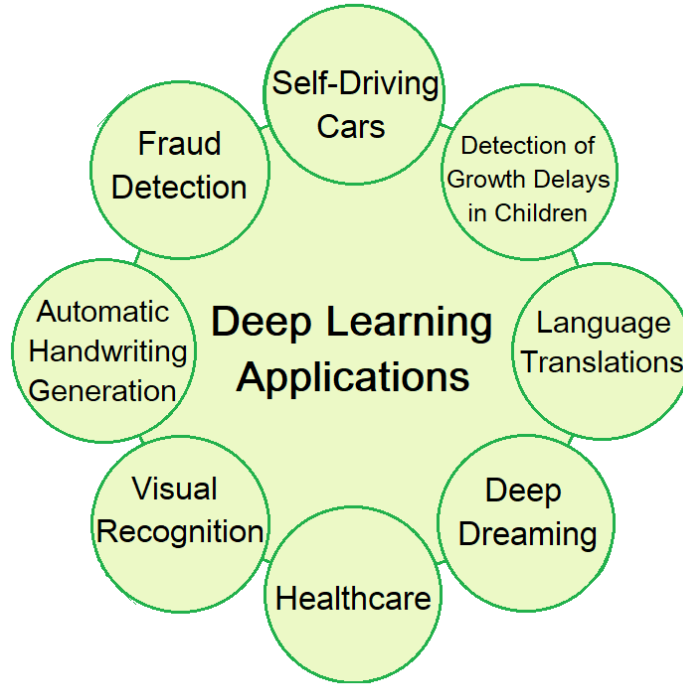


Figure 3-3 :Applications for Deep Learning

3.2.2 Types of Deep Learning Architectures

Different types of DL architectures are utilized by researchers and practitioners to build neural networks. These architectures are tailored to tasks and data types, and over time some models have emerged as fundamental in the field. In the below discussion, we will explore some of the deep learning architectures that have greatly contributed to the progress of AI capabilities.

a) Convolutional Neural Networks (CNNs):

CNNs are specifically designed for processing data with a grid-like topology, such as images. A CNN convolves learned filters over the input data, reducing the spatial size while increasing the depth of the feature maps, thereby learning hierarchies of features. This architecture excels at tasks like image and video recognition, image classification, and object detection (Hedayati et al., 2019; Nife & Chtourou, 2022; Sultana et al., 2018).

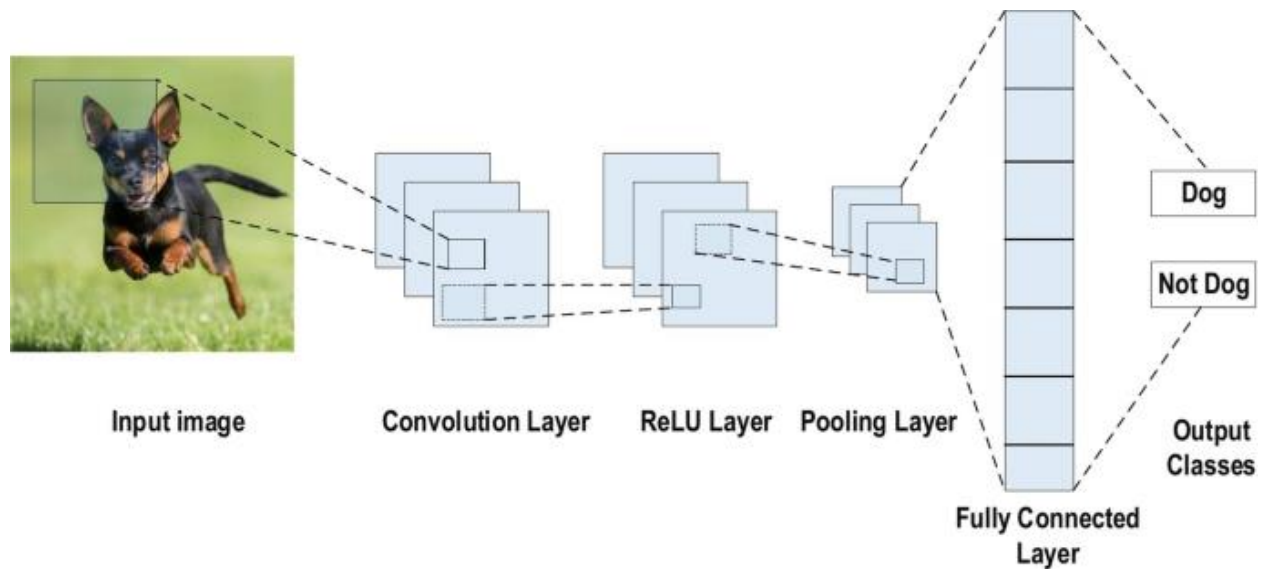


Figure 3-4: An Example of CNN architecture for Image Classification. Source: (Alzubaidi et al., 2021).

b) Recurrent Neural Networks (RNNs):

RNNs are suited for sequential data, such as time series or natural language. They have the distinctive feature of having feedback loops, allowing information to persist. In theory, RNNs can maintain information in 'memory' over time to process sequences of data. However, they are difficult to train over long sequences due to the vanishing gradient problem (J. Wang, 1994).

c) Long Short-Term Memory Networks (LSTMs):

LSTMs are an advancement of the RNN that can learn and remember over long sequences of data without the vanishing gradient issue. They achieve this with a complex system of 'gates' that regulate the flow of information into and out of the cell state, effectively allowing the network to maintain a longer memory (Iparraguirre-Villanueva et al., 2023).

d) AutoEncoders:

Autoencoders are unsupervised learning models that aim to learn compressed representations of the input data. They work by encoding the input into a lower-dimensional space and then decoding it back to reconstruct the input as closely as possible. Variants like Variational Autoencoders (VAEs) add probabilistic twist to this architecture and are used in generative tasks (J. Liu, 2023).

e) Generative Adversarial Networks (GANs):

GANs consist of two networks known as a generator and a discriminator. The primary function of the generator is to create data that closely resembles real data, while the discriminator's role is to assess whether the data is authentic or generated by the generator. This competitive process leads to the generation of remarkably high-quality data. GANs find applications in domains such as generating images, enhancing photographs, producing artistic content, and even creating realistic audio and video (Jieren et al., 2020).

f) Transformers:

Transformers have emerged as the architecture for handling data, particularly in natural language processing (NLP) tasks. Unlike RNNs and LSTMs, transformers enable parallel processing during training by not requiring sequential data to be processed in a specific order. They leverage on self-attention mechanisms to determine the importance of parts of the input data. Transformers have paved the way for models like Bidirectional Encoder Representations from Transformers (BERT) and Generative Pre-Trained Transformer

(GPT) that have raised performance standards across an array of NLP tasks (Kamath et al., 2022).

g) Capsule Networks (CapsNets):

Capsule networks are a type of neural network introduced by Geoffrey Hinton and his research team in 2017 (Sabour et al., 2017). CapsNets propose a new neural network structure that aims to overcome some of the limitations of CNNs, such as the need for large amounts of data and their inability to recognize spatial hierarchies between simple and complex objects. Capsules are groups of neurons that encode both the probability of the existence of a feature and its orientation.

h) Siamese Networks:

These networks are used for tasks that involve finding the similarity or relationship between two comparable things. They are especially useful in applications such as face recognition, signature verification, and other applications where it's critical to compare features of two inputs (Xiao & Ding, 2022).

3.3 Transfer Learning

Transfer learning is a powerful technique in machine learning where a model developed for a particular task is reused as the starting point for a model on a second task. It is a popular approach in deep learning that can train deep neural networks with comparatively less data. This is particularly useful in the field of deep learning where enormous datasets and considerable computational power are usually required to train deep learning models from scratch (Li, 2018). The essence of transfer learning lies in leveraging the knowledge that a model has gained from a

related task to improve its performance on another. It is particularly effective in situations where the target task has insufficient data to train a full-scale model effectively.

3.3.1 Process of Transfer Learning in deep learning:

a) Pre-training: A neural network is first trained on a large, often generic dataset. For example, models may be trained on ImageNet, which contains millions of labeled images across thousands of categories. During this phase, the network learns a wide range of features from the data.

b) Base Model for Feature Extraction: The pre-trained model is used as a feature extractor. The last few layers of the model are removed, and the remaining network is treated as a fixed feature extractor for the new data.

c) Fine-tuning: Sometimes, more than just the last few layers are adjusted or trained from scratch. The earlier layers may be frozen to retain the learned features, and the latter layers are retrained with a new dataset. In some cases, the entire network may be fine-tuned to better adapt to the new task.

d) Task-specific Adjustment: The model is then further adjusted to perform the specific task it's being repurposed for. This often involves adding new layers tailored to the new dataset or problem, such as new fully connected layers for classification tasks or new convolutional layers if additional feature extraction is necessary.

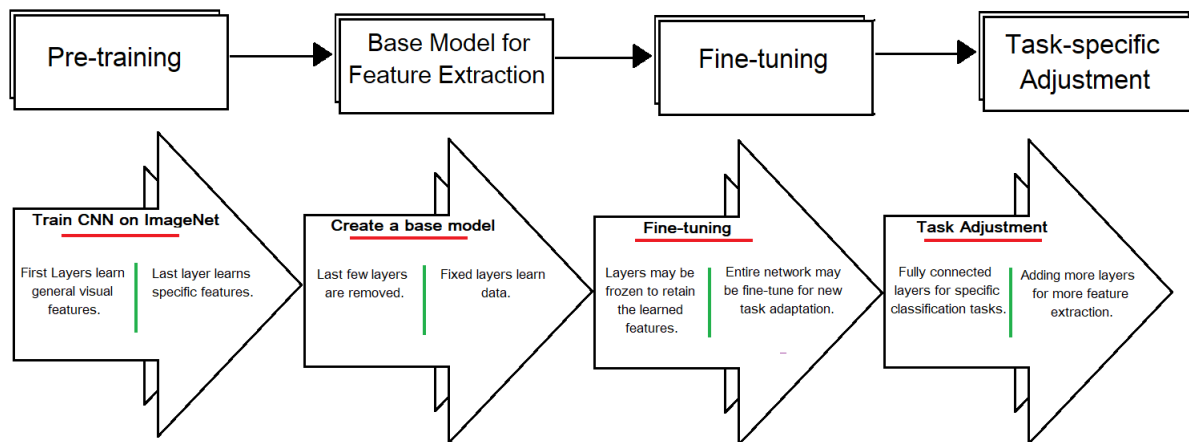


Figure 3-5: Transfer Learning Process

3.3.2 Advantages of Transfer Learning:

a) Reduced Dependency on Data: One major advantage of transfer learning is its ability to minimize the reliance on an amount of training data. By leveraging a trained model that has already learned relevant features on comprehensive dataset, it becomes possible to achieve high performance with less data.

b) Efficiency in Time and Computation: Training a deep learning model from scratch can be time consuming and computationally expensive, often taking days or even weeks of computation on specialized hardware. Transfer learning allows the utilization of pre-trained models, which significantly reduces both training time and computational costs.

c) Enhanced Performance: Models that are pre trained on larger datasets often exhibit better performance, particularly when applied to smaller datasets, because the models have already acquired rich set of features from the dataset, providing a solid foundation for the new task. This can result in accuracy and better generalization when fine-tuned for a specific task.

d) Flexibility and Adaptability: Transfer learning offers flexibility in adapting a model trained for one task to another task with minimal modifications. This adaptability proves useful in fields where tasks are similar but not identical, enabling rapid prototyping and deployment of models across various domains.

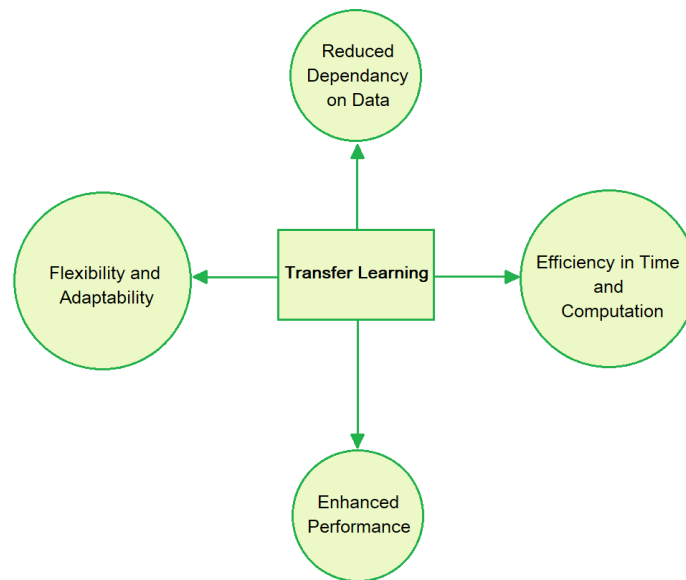


Figure 3-6: Advantages of Transfer Learning

3.4 Studied Models

3.4.1 ResNet50

ResNet50 belongs to the Residual Network (ResNet) family, which made an impact on the field of deep learning when it emerged as the winner of the ImageNet's Large Scale Visual Recognition Challenge (ILSVRC) in 2015 (He et al., 2016). The "50" in its name signifies that it consists of 50 layers in its architecture. ResNets were developed to address a problem known as the vanishing gradient issue, which often hindered training deep neural networks. This issue arises when the gradient of the loss function decreases exponentially with each layer making it challenging to train the network effectively.

Prior to ResNets introduction, deep neural networks faced difficulties as they grew in depth. Training such networks required resources and yielded fewer desirable results because adding more layers did not always lead to improved performance due to vanishing gradients. The groundbreaking approach of ResNet introduced "skip connections" or "shortcut connections," which allowed gradients to bypass one or more layers. These connections enable identity mapping by adding their outputs to those of the stacked layers. The core idea behind this approach is for these layers to learn functions in relation to their inputs, than learning functions with no reference point. This simplifies the learning process significantly.

layer name	output size	18-layer	34-layer	50-layer	101-layer	152-layer
conv1	112×112	7×7, 64, stride 2				
		3×3 max pool, stride 2				
conv2_x	56×56	$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$
conv3_x	28×28	$\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 8$
conv4_x	14×14	$\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 23$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 36$
conv5_x	7×7	$\begin{bmatrix} 3 \times 3, 512 \\ 3 \times 3, 512 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 512 \\ 3 \times 3, 512 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$
	1×1	average pool, 1000-d fc, softmax				
FLOPs		1.8×10^9	3.6×10^9	3.8×10^9	7.6×10^9	11.3×10^9

Figure 3-7: Versions of the ResNet architecture with varying depths. Source: (He et al., 2016)

3.4.1.1 Overview

ResNet50's architecture is built using a CNN backbone, which is fundamentally a stack of convolutional, batch normalization, and Rectified Linear Unit (ReLU) activation layers followed by a pooling layer and a Fully Connected (FC) layer at the end. A key feature of ResNet is its "residual block," which includes convolutional layers and a shortcut connection that bypasses these layers. In ResNet50, there are 16 residual blocks, each comprising three layers of

convolution. ResNet50 and other ResNet models have gained popularity in the field of deep learning due to their simplicity and effectiveness. They are widely used for tasks such as image classification, object detection and serving as backbones for complex models in various applications, like medical image analysis, and autonomous driving systems.

One significant advantage of ResNet50 is its ability to be trained with fewer parameters compared to other deep networks, while avoiding overfitting. The success and influence of ResNet50 highlight the significance of architectural innovations in DL. This introduction paved the way for research into residual learning and has inspired different advancements in neural network design.

3.4.2 VGG19

VGG19 is a deep convolutional neural network architecture named after the Visual Graphics Group from Oxford that developed the model. It gained fame for its simplicity and depth during its submission to the ILSVRC competition in 2014. VGG19 primarily comprises 19 convolutional layers with learnable parameters. The VGG models, including VGG19, were among the first to push the depth of convolutional neural networks beyond what had been seen before. What set VGG19 apart was its demonstration that network depth plays a major role in achieving high performance in large scale image recognition tasks.

3.4.2.1 Overview

VGG19s architecture stands out for its utilization of large number of convolutional layers with small (3x3) filters. These layers are stacked on top of each other with increasing depth. Reducing the size of the filters allows for deeper network without increasing the number of parameters. This is because using stacked filters mimics the impact of larger receptive fields

such as 5x5 or 7x7 filters, while requiring fewer parameters and hence better control over overfitting. The network utilizes convolution layers with ReLU activations, followed by max pooling layers. As the network deepens it progressively increases the number of filters. After several blocks of these convolution layers, the network ends with a few fully connected layers before the final classification layer, which uses the SoftMax activation function.

The simplicity of VGG19's architecture makes it highly accessible and comprehensible, contributing to its adoption. It has become a choice for transfer learning applications, where the network trained on a dataset like ImageNet is fine-tuned for a specific task with a smaller set of images. Despite its deeper structure, one prominent drawback of VGG19 is that, it requires more parameters compared to counterparts model like ResNet50 (Lai et al., 2023). This implies that training and inference with VGG19 can be computationally intensive and more prone to overfitting if not properly regularized and augmented with data. Nonetheless, the effectiveness of VGG19 in extracting image features has established it as a prominent tool in the fields of ML and computer vision. Its contribution to understanding of neural network architecture and its influence on subsequent designs remains substantial. The model's architecture continues to guide and inspire the development of new models in expanding network depth without compromising its ability to learn from and generalize across data.

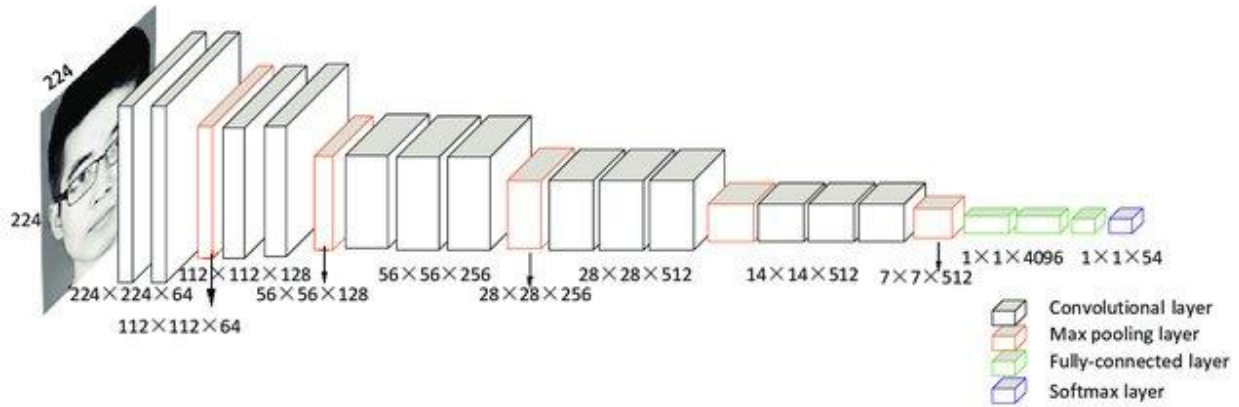


Figure 3-8: An illustration of the architecture of the VGG-16 model. Source: (Pei et al., 2019).

3.4.3 MobileNet

MobileNet is a neural network architecture specifically designed for devices with limited computational capabilities, such as smartphones and tablets. Its primary goal is to strike a balance between computational resource usage and model performance, enabling utilization of DL models on mobile devices. The MobileNet architecture was created by researchers at Google and introduced through a series of papers starting from 2017. This model was groundbreaking because it addressed the increasing demand for running these models on devices without relying on server communication (Howard et al., 2017).

3.4.3.1 Overview

MobileNet stands out for its efficiency, which is achieved through the utilization of depthwise separable convolutions. These types of factorized convolutions break down the operation into two layers; depthwise and pointwise convolutions. This factorization significantly reduces the computational costs and model size by separating the learning of spatial features from channel wise features. The core element of MobileNet building block lies in the depthwise separable convolution. In this process, a single filter is applied to each input channel (input depth) during depthwise convolution. Subsequently, a simple 1x1 convolution, known as

pointwise convolution, is used to combine the outputs of the layer. This combination of depthwise and pointwise convolutions forms a depthwise separable filter that greatly reduces the number of calculations compared to standard convolutions, resulting in computational resource savings.

In addition to its use of depthwise convolutions, MobileNet incorporates efficient features like width multiplier (Kim et al., 2023) and resolution multiplier (Howard et al., 2017) further enhancing its overall efficiency. MobileNets advantage lies not in having fewer parameters, but also requiring fewer calculations for forward passes through the network. This makes it highly suitable for real time applications on mobile devices such as facial recognition, object detection, and augmented reality where rapid and responsive models are crucial.

Despite its smaller size and lower computational requirements, MobileNet performs well on various benchmark tasks. This showcases the model's ability to maintain a high level of accuracy when working with limited resources. As a result, it has become a popular choice among researchers and developers who wish to deploy DL models in mobile applications that have constrained computational power. The impact of MobileNet has been significant, leading to the development of models and architectures that prioritize efficiency and speed. It continues to serve as a foundational model for those interested in exploring mobile and edge computing within the field of ML.

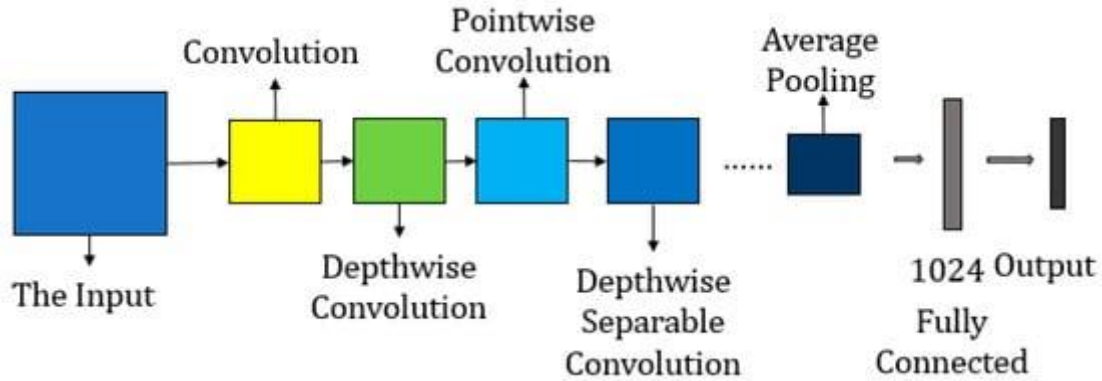


Figure 3-9: The structure of the MobileNet V1 network. Source: (Han et al., 2022)

3.5 Library and Tools

3.5.1 Keras and TensorFlow

Keras is an open-source software library that acts as an interface for TensorFlow library. It was developed with the aim of enabling fast experimentation. Keras is a high-level neural networks Application Programming Interface (API) that focuses on being user friendly, modular, and extensible. François Chollet originally created it as part of the ONEIROs (Open ended Neuro Electronic Intelligent Robot Operating System) research project (Chollet et al., 2015). The main objective of Keras is to facilitate the easy development of deep learning models. At its core, Keras allows for effortless and rapid prototyping of deep learning models. It seamlessly runs on both CPUs and GPUs, supporting convolution networks and recurrent networks, as well as combinations of the two types. Additionally, Keras also has support for arbitrary network architectures enabling users to create completely customized and innovative deep learning structures that are tailored to their specific problems.

Keras integrates smoothly with TensorFlow, which means it can take full advantage of TensorFlow's rich feature set. This includes manipulation of tensors, automatic differentiation, capabilities, and deployment on various platforms such as mobile devices or distributed servers.

By using Keras, TensorFlow becomes more accessible without compromising flexibility and performance. In terms of deep learning models specifically, Keras provides built-in support for data preprocessing tasks that are crucial for handling image and text data. It also provides tools that assist in building, training, and validating neural networks. These tools include structures and several pre trained models that can be utilized for transfer learning, fine tuning, or as feature extraction systems (Rimal, 2020).

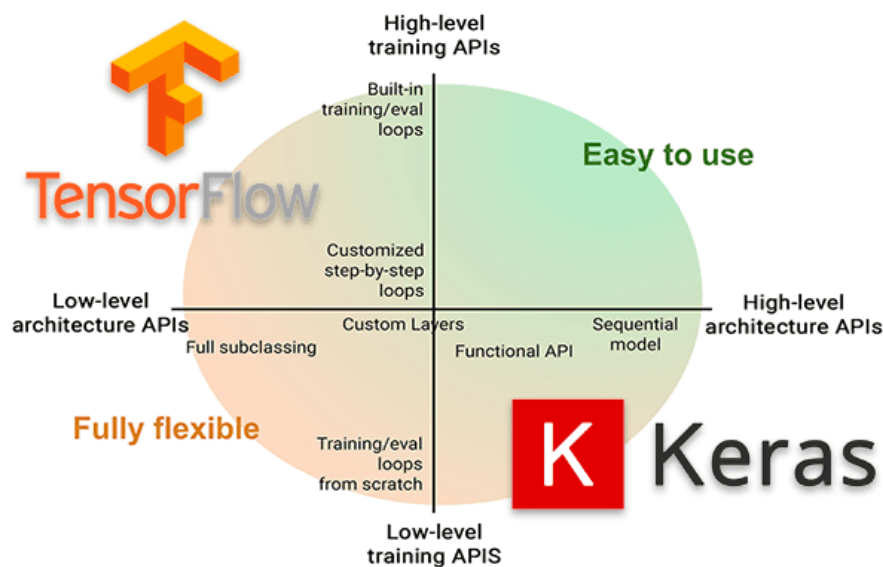


Figure 3-10: Keras and TensorFlow: Source: PyImageSearch

3.5.2 Python

Python has become widely popular in the ML community because of its easy-to-understand syntax, along with its collection of libraries. When it comes to data science and ML, many people prefer Python because it's user friendly and allows for model prototyping and iterative testing. The language is equipped with libraries like TensorFlow, PyTorch and scikit-learn, which are crucial for building and refining ML algorithms. Its intuitive syntax makes the development process faster. Additionally, Python's large and active community adds to its

appeal by providing a wealth of learning resources, continuous improvements, and a wide range of tools that make it an essential choice for creating sophisticated machine learning applications.

3.5.3 OpenCV

Open-Source Computer Vision (OpenCV), which is well known for its expertise in computer vision tasks is a library that offers a range of tools and algorithms for processing both images and videos. It not only allows for enhancing and manipulating images, but it also provides advanced functionalities like feature detection, object recognition, and even DL frameworks for complex tasks such as image categorization and object identification. Its flexibility makes it compatible with many programming languages, especially with Python, making it a top selection among developers in the field of computer vision.

3.5.4 Matplotlib

Matplotlib is widely regarded as a leading visualization library in the Python ecosystem. It empowers users to create an array of static, animated, and interactive graphics. What sets Matplotlib apart is its user-friendly interface that simplifies the process of generating plots, histograms, scatter diagrams and other visual representations. This makes it an essential tool for data analysis and exploratory research by transforming data into illustrative formats.

3.6 ImageNet

ImageNet is a large-scale dataset with annotations specifically designed for research on visual object recognition software. The project has diligently annotated over 14 million images, indicating the objects in them. Additionally, bounding boxes have been provided for one million of these images. ImageNet contains more than 20,000 categories with a typical category, such as "cat" or "dog", consisting of several hundred images. The motivation behind creating ImageNet

stemmed from the necessity for an extensive database of images to advance computer vision research. Prior to ImageNet's existence, researchers had access to small datasets that couldn't adequately support the development of algorithms requiring large scale and real-world data. Dr. Fei Fei Li, formerly associated with Princeton University took the lead in developing ImageNet (Deng et al., 2009). The project aimed to leverage crowdsourcing and the internet's capabilities to exponentially increase the creation of an annotated image dataset.



Figure 3-11: Example of an ImageNet dataset. Source: (Andrej, n.d.)

The structure of ImageNet draws inspiration from WordNet's organization that groups English words into sets called synsets representing synonyms (Gonzalo et al., 1998). Each synset

corresponds to a concept and ImageNet strived to provide an average of 500 - 1000 images for each synset — an objective that has been largely accomplished. The project utilizes workers from Amazon Mechanical Turk to annotate images with their corresponding synset, and to verify the relevance and quality of these images (Recht et al., 2019).

3.6.1 ImageNet Importance in Machine Learning

ImageNet has gained recognition for hosting the ImageNet ILSVRC, an annual competition where research teams evaluate their algorithms using the provided dataset and compete to achieve higher accuracy rates (Russakovsky et al., 2015). This competition has played a role in advancing the field of computer vision. ILSVRC has served as a benchmark for various DL models and architectures, particularly CNNs. In the 2012 edition of this challenge, AlexNets performance surpassed the traditional computer vision methods by a margin, marking the advent of the DL era in computer vision (Krizhevsky et al., 2017). Models that have excelled in the ImageNet challenge such as AlexNet, VGG, Inception (GoogleNet), ResNet and others, have become widely recognized frameworks for numerous computer vision tasks beyond classification. These tasks include object detection, semantic segmentation, and more.

Additionally, pre-trained models on ImageNet have been extensively embraced for transfer learning purposes where a model trained on the ImageNet dataset serves as a starting point for training on computer vision tasks. The reason for this is because ImageNet is very comprehensive, covering a variety of objects and situations, making it a valuable resource for machine learning models to learn from.

3.6.2 ImageNet Today.

ImageNet's impact on the development of ML and AI has been significant. It has demonstrated the importance of having large-scale, diverse, and well annotated datasets, to advance technology in these fields. ImageNet remains an essential foundation in computer vision, frequently used for training models and as a benchmark for new algorithms. As technology progresses, ImageNet continues to evolve with images and annotations that encompass a range of objects and concepts adapting to the expanding scope of ML applications.

Chapter 4: Research Methodology

4.1 Data Collection: The Cornerstone of Machine Learning

In ML, especially for applications as nuanced and complex as facial recognition, data serves as the foundational layer upon which the entire architecture is built. It's not just the quantity but also the quality and diversity of the data that determine the reliability, accuracy, and robustness of the final machine learning model. Poor or biased data can lead to misclassifications, reinforcing stereotypes, or simply inaccurate predictions that could significantly misguide educators in assessing student engagement and comprehension. Therefore, painstaking effort and consideration were invested in data collection for this research, ensuring that the dataset was rich, balanced, and reflective of the complexities involved in facial expression recognition.

Recognizing the intricate nature of facial expressions and the challenges they present in educational settings; we collected data from two different sources: a public dataset called AffectNet and a dataset that was captured by our research team to mimic a VR-based classroom setting.

4.2 Online Dataset: AffectNet

AffectNet is a large-scale, publicly available dataset specifically designed for facial expression recognition in the wild, meaning the images are not confined to controlled environments but reflect real-world conditions. It contains over one million facial images collected from the Internet, annotated for eleven different facial expressions: happiness, sadness, surprise, neutral etc. (Figure 2) The strengths of using AffectNet as a source include its large sample size and diversity, encompassing different ages, genders, and ethnicities, thereby making

the model more generalizable (Mollahosseini et al., 2019). However, it's important to note that while AffectNet provides a broad scope, it may not be specifically tailored for educational settings, especially VR-based educational settings. For example, confusion expressions which are important for our study is missing in AffectNet dataset. We have therefore collected a second dataset.



Figure 4-1: Sample images from the AffectNet dataset, showing different emotions: neutral, happiness, and boredom, respectively.

4.3 Custom Dataset and its Collection

To address the limitations of the AffectNet dataset and to make our model more attuned to the specificities of a VR-based educational environment, a second dataset was captured by our research team.

4.3.1 Ethics Application: The Necessity of Ethical Conduct in Human-Centric Research

Conducting research that involves human participants demands strict ethical considerations to ensure the dignity, rights, and welfare of the subjects involved. Receiving ethics approval lends credibility to the research and provides participants with the assurance that their well-being has been duly considered.

In light of this, an ethics application was submitted to the University of Calgary Conjoint Faculties Research Ethics Board (REB22-0832). After fulfilling all stipulated requirements, the application was approved within a two-month period, thus paving the way for the commencement of data collection.

4.3.2 User Study Design and Accessibility

Our user study materials were meticulously designed to be accessible and straightforward, ensuring that participants could easily follow instructions. The materials were in the form of a video presentation, which guided the participants through the data collection process.

To create a user-friendly experience, the video utilized text-to-speech instructions generated through lovo.ai directing participants on the required facial expressions. Recognizing the diversity of our participant pool and to cater to those with hearing impairments, subtitles were added using the 'VideoProc Vlogger' software. The video commenced by welcoming participants and briefly explaining the objective of the study. It also provided a preparation period for the participants to ready themselves for the recording sessions. Each facial expression was timed for 5 seconds, introduced by a countdown to help participants prepare.

To facilitate natural and authentic facial expressions, specific background visuals and music tracks were chosen for each expression type. For example, a cheerful background coupled with the lively tune "Sunny Over The Horizon" by DJ Freedom was used to elicit happiness expression. Conversely, a plain, dull background along with sleep-inducing music was selected to evoke expressions of boredom, which ensured genuine expressions and an engaging user experience.

4.3.3 Recruitment Process

One of the critical aspects of any study involving human subjects is the recruitment of participants. Our target demographic for this research comprised graduate students from the Computer Science department at our institution - University of Calgary, contacted through a dedicated mailing list. The recruitment electronic mail (Email) introduced the study's objectives, an e-poster, participation incentives and expectations, thus allowing them to make an informed decision about taking part in the study.

4.3.4 Diversity and Inclusivity

The recruitment was designed to be as inclusive as possible, inviting participants of all races, genders, and age groups. This broad-based approach aimed to ensure that our dataset was not limited to a particular demographic but rather was as diverse and representative as possible. We successfully recruited a total of 30 participants, thus creating a rich dataset for our facial expression recognition model.

Table 4-1: Table showing participant's details.

Participant Number	Age Group	Gender	Ethnicity
5, 28	25, 27	M, F	East Asian
6, 7, 12, 25, 26, 27	29, 34, 25, 27, 24, 32	F, F, M, F, F, F	Black
15	26	F	Middle Eastern
1, 3, 11, 17, 20, 21, 23, 29	29, 26, 25, 26, 28, 28, 25, 28	M, M, M, M, F, M, M, F	South Asian
2, 4, 8, 9, 10, 13, 14, 16, 18, 19, 22	26, 24, 25, 26, 25, 28, 26, 24, 25, 33, 25	M, M, Non-Binary, M, M, F, M, F, M, M, PNTA	West Asian
24, 30	30, 25	M, F	Latino

We asked participants to self-report demographic information based on their age, gender, and ethnicity. Participants numbered 5 and 28 were aged 25 and 27 respectively, identified as male and female, and were of East Asian descent. Those labeled 6, 7, 12, 25, 26, and 27 were between the ages of 29 and 32, identified as male and female, and were of Black ethnicity. Participant 15 was a 26-year-old female from the Middle Eastern region. Participants 1, 3, 11, 17, 20, 21, 23, and 29 ranged in age from 25 to 29, with a gender distribution of six males and two females, and all were of South Asian origin. Participants 2, 4, 8, 9, 10, 13, and 14, aged between 24 and 26, comprised of five males, one female and a non-binary individual, all hailing from West Asia. Participants numbered 16, 18, and 19 spanned ages from 24 to 33 and had a gender split of two males and one female. Participant 22 was undisclosed - Prefer Not To Answer (PNTA), while participants 24 and 30, aged 30 and 25 respectively, were both male and female of Latino ethnicity."

4.3.5 User Study Setup: Creating a Controlled yet Natural Environment

AI research lab, chosen for its spacious and distraction-free environment, served as the venue for data collection. We configured the lab to meet the requirements of the study, focusing on both technical and comfort aspects. A computer was set up complete with an external speaker to play the instructional video to the participants. To capture high-quality facial expression data, a Nikon Professional camera was strategically positioned on a tripod behind the computer. The camera was aimed at the chair where participants would be seated, ensuring that their faces would be recorded from head to chest. A chair was placed in front of the computer to offer a comfortable position for the participant during the recording. The setup was deliberately designed to mimic a natural setting, focusing on recording detailed expressions vital for training

our machine learning models and ensuring participants could express genuine facial expressions, which are essential for the validity of our study.

4.3.6 Data Capturing Day:

4.3.6.1 Scheduling and Informed Consent

To ensure that the data capturing process ran smoothly, appointments were pre-scheduled with interested participants who responded to our initial email. Upon arrival at the lab, each participant was greeted warmly and provided with a consent form. This document outlined the research objectives and the potential risks involved in participation, including the unlikely but possible physical fatigue from displaying various facial expressions.

4.3.6.2 Participant Rights and Safeguards

Participants were clearly informed that they were free to withdraw from the study at any point, and if they chose to do so, all their recorded data would be deleted. This ensured that the participants could exercise their autonomy throughout the study, which is in line with ethical guidelines for human-subject research. The participants were then escorted to the prepared setup, featuring a chair in front of a computer with an attached external speaker and a camera on a tripod. Each participant was recorded individually to ensure the integrity and quality of the data, focusing on four facial expression classes: Neutral, Boredom, Confusion, and Happiness, which were targeted for this study.



Figure 4-2: User Study Setup

4.3.7 Data Processing: From Videos to Usable Data

We gathered data in a single, uninterrupted recording video format, capturing all four facial expressions per participant, resulting in 30 comprehensive videos. These were subsequently divided according to expression classes, totaling 120 videos. This step was crucial for segregating the data into different classes, aiding the supervised learning approach we adopted for training our ML models. They were recorded at 23.98 frames per second (fps) but were down sampled to 15 fps to reduce computational strain. After converting these videos into image frames, we reviewed and discarded unsuitable frames where the images were blurry, resulting in 9,000 usable frames.

4.3.8 Conversion to Grayscale

To enhance our dataset and improve the robustness of our model we decided to convert our RGB images to grayscale. This conversion helps emphasize shapes features than color features, and also reduces the computational load unlike RGB images that have three channels (Red, Green and Blue), grayscale images only have one channel. The standard method we used

for this conversion involves calculating the grayscale value for each pixel using a sum of the R, G and B components.

$$\text{Grayscale Value} = 0.299 \times R + 0.587 \times G + 0.114 \times B \quad (4-1)$$

The weights are determined based on how our eyes perceive colors. Our eyes are most sensitive to green light, followed by red light and least sensitive to blue light. Thus, when converting to grayscale, the green component gets the highest weight, and the blue component gets the lowest.

4.3.8.1 Significance of Grayscale Value

- a) Reduced Computational Burden:** Since grayscale images only have one channel there is a significant reduction in computations during training.
- b) Simplification:** By removing the color details from our datasets, we simplify the complexity of the images, making them easier for processing and analysis. This allows the model to become better at recognizing patterns, edges and shapes which are crucial for our work.
- c) Consistency and Uniformity:** When working with datasets that contain images from various sources, it's common to encounter inconsistencies in color properties caused by variations in lighting, camera quality or post processing techniques. Converting the images to grayscale helps establish a sense of uniformity across the dataset and ensures that all images are analyzed on a uniform scale.

4.3.9 Leveraging Facial Landmark Detection

A vital component in our data processing pipeline is the use of ``shape_predictor_68_face_landmarks`` from the “dlib” Python library. This pre-trained model serves as a facial landmark detector that estimates the positions of 68 points on the face (Figure 4-3). These landmarks correspond to various facial structures such as the eyes, nose, and mouth. It was developed initially for applications in facial recognition and augmented reality, the predictor has demonstrated high accuracy in identifying these facial features.

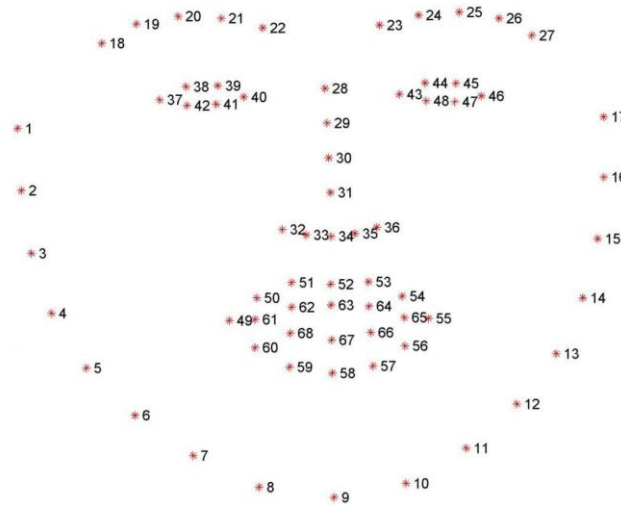


Figure 4-3: Facing feature identification from the dlib Python library, showing full face identifications. Source: (Sagonas et al., 2013).

4.3.9.1 Processing for Full Face Recognition

To target only facial region, we cropped bounding boxes around each face, specifically, we set point x to 20 and point y to 25, extracting only the full face from the frames that original looks like a passport-sized photos. This dataset serves as the foundation for our full-face recognition model, providing a robust set of features without irrelevant background or body information. This focuses the model's attention on the face alone, increasing the efficacy of the training process.

4.3.9.2 Processing for Facing Occlusion

Considering the relevance of VR in education, our research also focuses on predicting facial expressions when the upper face is obscured by a VR headset. Thus, we processed data to highlight the lower face, the area visible even when a VR headset is worn. Using different cropping parameters “point x at 3 and point y at 15” (Figure 4-4), we emphasized this lower region for our occlusion-specific model. Conclusively, all cropped images, both full and occluded half-face, were resized to 244x244 pixels, ensuring they fit the parameters required by our training model.

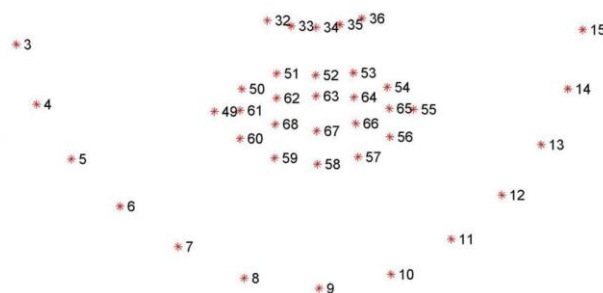


Figure 4-4: Facing feature identification from the dlib Python library, showing Half face identifications. Source:(Sagonas et al., 2013). (Edited to fit this research purpose)

4.4 Customized Training Approach ResNet50

In our quest to build a specialized DL model capable of distinguishing between nuanced facial expressions, particularly in the context of online education and VR classrooms, we opted for a customized training approach ResNet50 architecture. ResNet50, a CNN with 50 layers, has been proven to perform exceptionally well in image classification tasks. However, to better suit the specificities of our study, certain modifications were imperative.

4.4.1 Batch Settings, Epochs, and Random Seeds

Efficiency and reproducibility are two pillars of any scientific research. Towards that, several key parameters were set before training the model. After experimenting with different batch sizes, we found that a batch size of 16 yielded the best results in terms of accuracy and computational efficiency. All our models were trained for 100 epochs. This number was found to be sufficient for the model to converge without overfitting. Setting the random seed for both NumPy and TensorFlow to 42 ensures that the experiments are reproducible.

```
[ ] # General parameters
    batch_size = 16
    epochs = 100
    size = 64
    np.random.seed(42)
    tf.random.set_seed(42)
```

Figure 4-5: General Parameters

4.4.2 Data Loading and Preprocessing

We utilized Keras' ImageDataGenerator to automate the data augmentation and batching processes. Our training and validation dataset were loaded in (Figure 4-6). The training data is stored in a Google Drive folder, conveniently accessed via Colab. Images were resized to 64x64 pixels to maintain uniformity across the dataset. As experiments 1 to 3 are multi-class classifications, the *class_mode* was set to '**categorical**,' except for experiment 4, which is a two-class classification; here, our *class_mode* was set to '**binary**.'

```
[ ] train_dataset = train_datagen.flow_from_directory(directory='/content/drive/MyDrive/Experiment 1/GreyHalfFace/Train',
    target_size=(size,size),
    class_mode='categorical',
    batch_size=batch_size)
```

Figure 4-6: Training and Validation data directory

4.4.3 Data Augmentation

Data augmentation is an essential technique used to artificially enhance the diversity of a dataset without collecting new data. By introducing minor alterations to the original images, we can simulate the variability that may be encountered in real-world scenarios. This helps in improving the generalization capabilities of models, especially when the amount of training data is limited. During our training (Figure 4-7), we adopted the following data augmentation techniques using TensorFlow's Keras API:

a) Random Horizontal Flip: The `tf.keras.layers.RandomFlip('horizontal')` function randomly flips an image along its horizontal axis. This augmentation imitates the variance that could arise if the object or subject in the image were positioned differently, potentially facing the opposite direction. For example, if the original image depicted a face looking left, the augmented image might show the face looking right.

b) Random Rotation: With the `tf.keras.layers.RandomRotation(0.2)` function, each image in the dataset is randomly rotated by a factor of 0.2. This means the images can be rotated anywhere in the range of (-20%, 20%) of their original orientation. This is beneficial in scenarios where the subject's angle might vary, helping the model recognize features from different orientations.

```
✓ [26] data_augmentation = tf.keras.Sequential([  
0s   tf.keras.layers.RandomFlip('horizontal'),  
      tf.keras.layers.RandomRotation(0.2),  
  
      #rotation_range=5,  
      ])
```

Figure 4-7: Data Augmentation Technique

To visualize the effects of our data augmentation techniques, we took a sample image from the training dataset and applied the augmentations. Using a subplot structure (Figure 4-8), the original image and its eight augmented variants were displayed in a 3x3 grid. This visualization allowed for a direct comparison of the original image with its augmented versions, showcasing the subtle differences introduced by the data augmentation techniques (Figure 4-9). Such augmentations help in ensuring that our model is not only trained on the base data but also on its numerous variations. By doing so, we equip the model to handle a wide range of input scenarios, making it more robust and adaptable.



```
for image, _ in train_dataset.take(1):
    plt.figure(figsize=(10, 10))
    first_image = image[0]
    for i in range(9):
        ax = plt.subplot(3, 3, i + 1)
        augmented_image = data_augmentation(tf.expand_dims(first_image, 0))
        plt.imshow(augmented_image[0] / 255)
        plt.axis('off')
```

Figure 4-8: Subplot structure code to view our sample argumentation.



Figure 4-9: 3x3 grid visualization of our sample argumentation data

4.4.4 Model Architecture: ResNet50 Base

For the feature extraction part of our model, we leveraged the pre-trained ResNet50 architecture. We excluded the fully connected layers at the top of the pre-trained model (`include_top=False`) to adapt the model for our custom classification layer. The model used pre-trained weights from the ImageNet database for quicker and more effective training. Figure 4-10 shows the entirety of our customized training approach ResNet50 architecture.

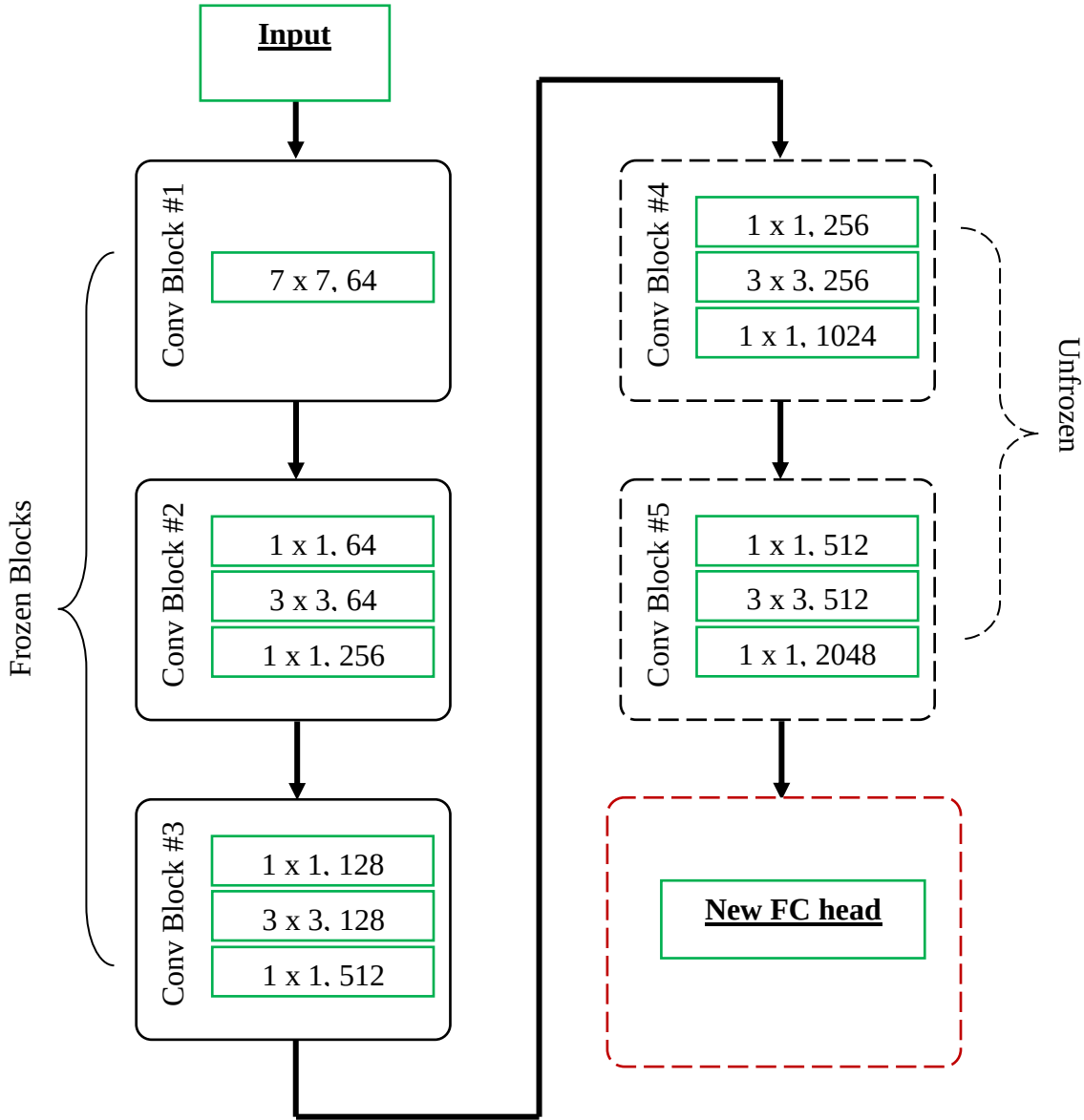


Figure 4-10 : Block diagram of the proposed customized ResNet50 architecture.

4.4.5 Layer Freezing Strategy

In transfer learning, it is often beneficial to freeze the weights of the pre-trained model during the initial phase of training. This ensures that the useful features captured by the pre-trained model are not distorted. We froze all layers except for the last one (`layers[:-1]`). This approach was taken to avoid overfitting, which becomes more likely as more layers are unfrozen and fine-tuned.

4.4.6 Unfreezing Specific Blocks

The ResNet50 architecture, inherently comprising multiple residual blocks, was further fine-tuned. Specifically, we unfroze the layers in blocks 4 and 5 for training, while keeping the preceding layers frozen. This localized training enabled the model to learn higher-level features pertinent to our specific use-case, without altering the generic feature-maps learned initially. The layer indices were programmatically set to be trainable starting “*fine_tune_at = 81, conv4_block1_1_conv*” (Figure 4-11). This method focused training on the deeper layers responsible for higher-level feature extraction, offering the dual benefits of time-efficiency and specialized learning.

```
#Fine-tune from this layer onwards
fine_tune_at = 81

#Freeze all the layers before the `fine_tune_at` layer
for layer in base_model.layers[fine_tune_at:]:
    layer.trainable = True
```

78	conv3_block4_3_bn	False
79	conv3_block4_add	False
80	conv3_block4_out	False
81	conv4_block1_1_conv	True
82	conv4_block1_1_bn	True
83	conv4_block1_1_relu	True
84	conv4_block1_2_conv	True

Figure 4-11: Unfreezing Blocks.

4.4.7 Sequential Model with Base Model Integration

The next step in customizing the training approach involved incorporating the base ResNet50 model into a sequential Keras model. This approach allowed us to append layers conveniently, thereby offering greater flexibility for architectural modification. The following code snippet in Figure 4-12 illustrates this process:

```
#Building Model
model = Sequential()
model.add(base_model)
```

Figure 4-12: Building Model


```
# Model Summary
model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
=====		
resnet50 (Functional)	(None, 2, 2, 2048)	23587712
dropout (Dropout)	(None, 2, 2, 2048)	0
flatten (Flatten)	(None, 8192)	0
batch_normalization (Batch Normalization)	(None, 8192)	32768
dense (Dense)	(None, 32)	262176
batch_normalization_1 (Batch Normalization)	(None, 32)	128
activation (Activation)	(None, 32)	0
dropout_1 (Dropout)	(None, 32)	0
dense_1 (Dense)	(None, 4)	132
=====		
Total params: 23,882,916		
Trainable params: 15,254,756		
Non-trainable params: 8,628,160		
=====		

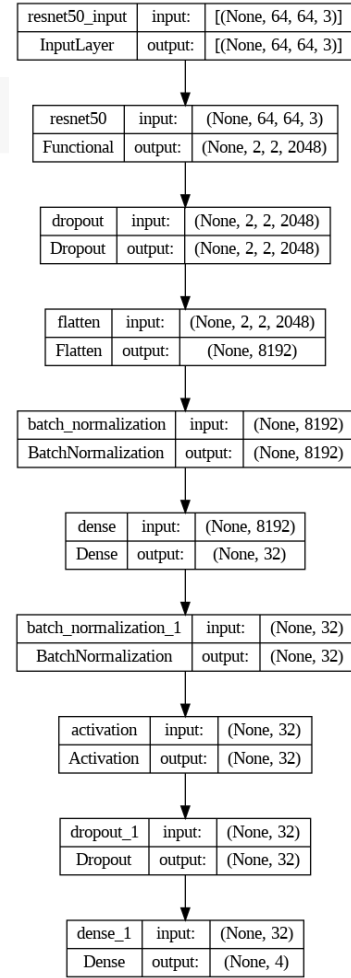


Figure 4-13: Customized Training Approach ResNet50 Model Summary.

4.4.8 The Challenge of Overfitting

One of the most daunting challenges in machine learning, particularly in DL models like ours, is overfitting. Overfitting occurs when a model learns the training data too well, capturing noise in addition to the underlying patterns. This over-specialization on the training data leads to poor performance on new, unseen data. Overfitting is especially a concern in our study, given the high dimensionality of facial images and the nuances of facial expressions.

4.4.9 Regularization and Dropout for Overfitting Control

To mitigate overfitting, dropout layers were strategically placed in our neural network, following both the base ResNet50 model and the subsequent dense layer. In each dropout layer, a percentage of the layer's input units are randomly set to zero during training, which helps to prevent any single neuron from becoming overly specialized. Specifically, we used a dropout rate of 0.5 following the base model and 0.6 after the dense layer (Figure 4-14). These rates were determined experimentally to maximize model generalization without significantly compromising training accuracy.

4.4.10 L2 Regulation

Additionally, L2 regularization was applied to the dense layers in the model. Regularization techniques like L2 are commonly employed to constrain the optimization process, discouraging the learning algorithm from fitting the training data too closely. L2 regularization adds a penalty term to the loss function based on the magnitude of the layer weights. In our case, we used a regularization factor of 0.3, which was empirically found to be effective in reducing overfit.

4.4.11 Customizing Output Layers for Different Experiments

To adapt our model to various experimental scenarios, different output layers were used:

- a) Experiment 1 and 3:** The output layer consists of four neurons, each corresponding to one of the primary facial expressions targeted in our study: Neutral, Boredom, Happiness, and Confusion. A soft argmax (SoftMax) activation function was employed to convert the raw output scores into probabilities, making it ideal for this multi-class classification problem.

- b) Experiment 2:** For this experiment, the output layer was tailored to classify only three primary facial expressions: Neutral, Happiness, and Confusion. Again, the 'SoftMax' activation function was used for probability conversion.
- c) Experiment 4:** Unlike the other experiments, here the output layer features only two neurons. These neurons correspond to two primary facial expressions targeted in the experiment. For this binary classification problem, we used the 'sigmoid' activation function, which is more suitable for a two-class setup.

```
#Building Model
model = Sequential()
model.add(base_model)

model.add(Dropout(0.5))
model.add(Flatten())
model.add(BatchNormalization())

model.add(Dense(32, kernel_initializer='he_uniform', kernel_regularizer=l2(0.3)))
model.add(BatchNormalization())
model.add(Activation('relu'))
model.add(Dropout(0.6))

model.add(Dense(3, activation='softmax'))
```

Figure 4-14: Regularization, Building Model with three neurons output layer.

4.5 Model Optimization

Building a DL model involves more than just architecture design; it's equally crucial to optimize the training process for optimum performance. We adopted several advanced techniques to improve training efficiency, prevent overfitting, and expedite convergence. Below, we detailed these methodologies, each serving a unique purpose in our model training.

4.5.1 Learning Rate Scheduling: ReduceLROnPlateau

Learning rate plays a critical role in the convergence of a model. Too high a rate may cause overshooting, while a low rate may result in slow convergence. To manage this, we used Keras' "*ReduceLROnPlateau*" callback (Figure 4-15). This function dynamically adjusts the

learning rate whenever the validation loss plateaus, thus speeding up the process and also potentially improving the final model.

```
lrd = ReduceLROnPlateau(monitor = 'val_loss',patience = 5, verbose = 1,factor = 0.50, min_lr = 1e-4)

mcp = ModelCheckpoint('model.h5')

es = EarlyStopping(monitor='val_loss', mode='min', verbose=1, patience=10)

optimizer = Adam(learning_rate=0.0001)
```

Figure 4-15: ReduceLROnPlateau callback

Here (Figure 4-15), “*monitor=val_loss*” specifies that the learning rate should be adjusted based on the validation loss. The “*patience=5*” parameter means that the learning rate will be reduced if no improvement in validation loss is observed for 5 epochs. The “*factor=0.50*” parameter halves the learning rate, and “*min_lr = 1e-4*” sets a lower limit to how much it can be reduced.

4.5.2 Model Checkpointing

During training, it's vital to save the model weights that provide the best validation performance. We used Keras' “*ModelCheckpoint*” (Figure 4-16) for this purpose. This ensures that the model weights are saved in a Hierarchical Data Format version 5 (HDF5) file format after every epoch, preserving not only the model architecture but also optimizer states. This facilitates resuming training if needed.

```
mcp = ModelCheckpoint('model.h5')
```

Figure 4-16: Model Checkpoint

4.5.3 Early Stopping

To prevent overfitting and reduce computational waste, we employed early stopping. This mechanism halts training when the monitored metric, in our case, “*val_loss*”, stops improving for a specified number of epochs “*patience=10*”. We used the Adam optimizer with a learning rate of “*0.0001*”, which is known for its effective adaptiveness in learning rates across parameters.

4.5.4 Model Fitting and Training Callbacks

Finally, the model was trained using the “*fit*” method, which was configured to use a set of callbacks for learning rate reduction, early stopping, and Weights and Biases (Wandb) for experiment tracking. This final step (Figure 4-17) amalgamated all the preceding techniques to ensure an efficient and effective training regimen.

```
[ ] history=model.fit(train_dataset,validation_data=valid_dataset,epochs = 100, verbose = 1, callbacks=[lrd,es,WandbCallback()])
```

Figure 4-17: Model Fitting and Training Callbacks

Chapter 5: Experimental Results.

5.1 Evaluation Metrics

Evaluation metrics are crucial for assessing the performance of machine learning models, particularly in classification tasks where the prediction of categories or classes is involved. These metrics provide a quantitative basis for comparing models, understanding their strengths and weaknesses, and guiding the improvement of algorithms.

5.1.1 Confusion Matrix

The foundation of many evaluation metrics is the confusion matrix, a table that is used to describe the performance of a classification model on a set of data for which the true values are known. It contrasts the actual labels with the model's predictions, categorizing them into four types of outcomes:

a) True Positives (TP): The cases in which the model correctly predicts the positive class.

Example: The facial recognition system is presented with an image of a person showing happiness. The model correctly identifies and classifies the image as 'Happiness'.

b) True Negatives (TN): The cases in which the model correctly predicts the negative class.

Example: A true negative for 'Happiness' mean that the model correctly identifies that the person is not happy in any given image that does not show happiness.

c) **False Positives (FP):** The cases in which the model incorrectly predicts the positive class, also known as a Type I error.

Example: The model is given an image of a person with a neutral expression, but it incorrectly classifies the expression as 'Happiness'. For the 'Happiness' class, this would be a false positive.

d) **False Negatives (FN):** The cases in which the model incorrectly predicts the negative class, also known as a Type II error.

Example: The system fails to identify 'Happiness' when it is present in an image, instead classifying the expression as 'Neutral'. This would be a false negative for the 'Happiness' class.

		PREDICTED LABEL	
		NEGATIVE	POSITIVE
TRUE LABEL	NEGATIVE	TRUE NEGATIVE	FALSE POSITIVE
	POSITIVE	FALSE NEGATIVE	TRUE POSITIVE

Figure 5-1: Confusion Matrix

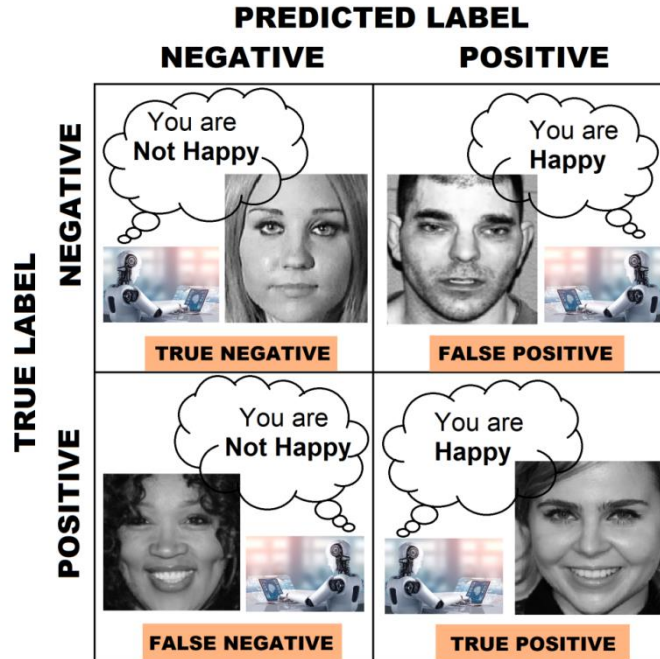


Figure 5-2: Confusion Matrix Scenario for an AI Predicting Happiness vs. Neutral

5.1.2 Derived Metrics from the Confusion Matrix

a) Accuracy: Accuracy is the most intuitive performance measure, and it is simply a ratio of correctly predicted observation to the total observations. It answers the question: "Out of all the classifications, how many did the model classify correctly?" However, accuracy alone is not a reliable metric in cases where the data is unbalanced (i.e., one class is much more frequent than others). (Eq. 5-1).

$$Accuracy = \frac{TN + TP}{TN + FP + TP + FN} \quad (5-1)$$

b) Precision: Precision is the ratio of correctly predicted positive observations to the total predicted positive observations. It is a measure of a classifier's exactness. The best value is 1 and the worst value is 0. (Eq. 5-2).

$$Precision = \frac{TP}{TP + FP} \quad (5-2)$$

- c) **Recall:** Recall is the ratio of correctly predicted positive observations to the all observations in actual class - yes. It is a measure of a classifier's completeness. The best value is 1 and the worst value is 0. (Eq. 5-3).

$$Recall = \frac{TP}{TP + FN} \quad (5-3)$$

- d) **F1 Score:** The F1 Score is the weighted average of Precision and Recall. Therefore, this score takes both false positives and false negatives into account. It is a measure of the test's accuracy. (Eq. 5-4).

$$F1 \text{ Score} = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (5-4)$$

5.2 Experiments and Results

Our experiments can be broadly categorized into two segments. The first segment is dedicated to the training of different neural networks on dataset. Following the training process, we saved the learned weights of each model for use in the second segment, which focuses on prediction tasks using the model unseen datasets. The four set of experiments are summarized in Table 5-1.

5.2.1 Experiment 1: Comparative Analysis of ResNet50, VGG19, and MobileNet

The goal of this experiment is to determine the effective neural network architecture for transfer learning in our specific facial recognition research. We chose common image classification architectures for comparisons based on recent works(Jahnavi et al., 2023;

Mascarenhas & Agarwal, 2021; Mohapatra et al., 2021; Victor Ikechukwu et al., 2021). Although there are performance statistics for ResNet50, VGG19, which is an updated architecture of VGG, and MobileNet, these existing metrics are general and do not accurately represent how these models perform in the specific scenario of emotion detection in VR-occluded faces. Hence, we conducted this comparative analysis. All the Tables and Figures in this segment of our experiment 1 show the results.

Table 5-1: Summary of our Experiments.

Experiments	Model Comparison	Dataset Used	Facial Expression Classes	Dataset for Comparison
Experiment 1	Default ResNet50 vs VGG19 & MobileNet	Our specialized dataset (training and prediction)	Neutral, Boredom, Confused & Happiness	Full Face and Half Face
Experiment 2	Default ResNet50 vs Custom-ResNet50	AffectNet (training) + Our specialized dataset (prediction)	Neutral, Boredom & Happiness	Full Face and Half Face
Experiment 3	Default ResNet50 vs Custom-ResNet50	Our specialized dataset	Neutral, Boredom, Confused & Happiness	Full Face and Half Face
Experiment 4	Custom-ResNet50	Our specialized dataset (training and prediction)	Boredom vs Happiness Boredom vs Neutral Boredom vs Confused.	Full Face and Half Face

5.2.1.1 Selection Criteria

a) ResNet50: We chose ResNet50 because of its depth and the presence of residual connections, which have been proven to aid in solving the vanishing/exploding gradient problem in deep networks. Additionally, ResNet50 is widely adopted in the research community and has been cited in numerous papers.

b) VGG19: We selected VGG19 based on its reputation for performance in image classification tasks and its straightforward architecture, which makes it easier to understand and modify.

c) MobileNet: Was included in our experiment because it serves as the architecture for Google's Teachable Machine (Carney et al., 2020). This architecture is specifically designed to be lightweight and efficient, making it highly suitable for real time applications or situations where computational resources are limited.

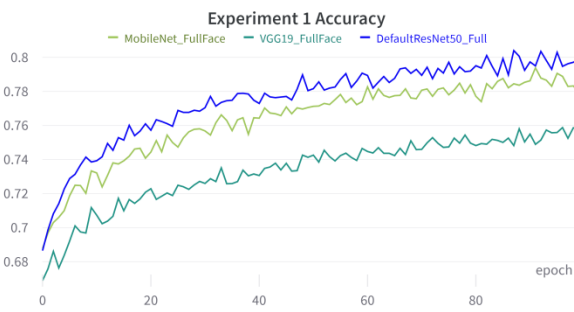
5.2.1.2 Training Parameters and Configuration

All three networks were trained using our dataset consisting of four classes: “Neutral”, “Boredom”, “Confusion”, and “Happiness”. We conducted the training over 100 epochs to ensure that the model reached a level of convergence. We also extended our experiment to account for the occluded “half-face” scenario, simulating the conditions under which a portion of the face is obscured when a student is wearing a VR headset.

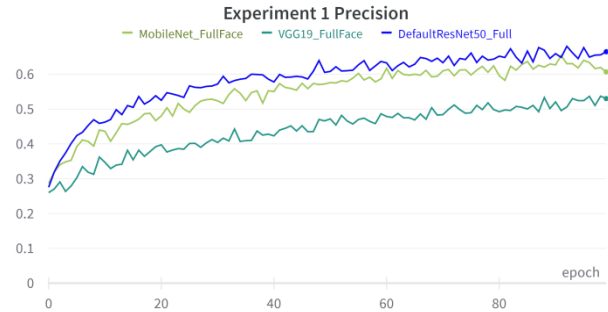
The results indicated a clear preference for the ResNet50 architecture in terms of both accuracy and generalization when applied to our dataset. This was consistent across both the full-face and occluded half-face conditions. This justifies our choice in using ResNet50 as the base architecture for our customization. Detailed metrics are presented in (Table 5-2), encapsulating various performance indicators such as accuracy, precision, recall, F1-score, etc. for a comprehensive view.

Table 5-2: Experiment 1 Training Results. (Full Face = F, Half Face = H)

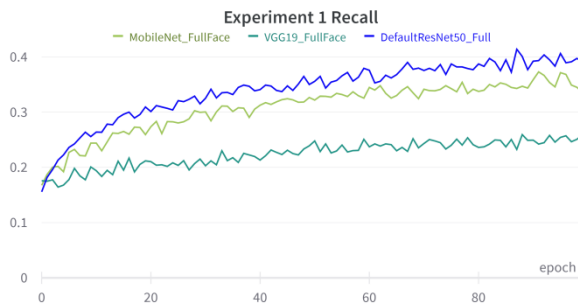
Model	Accuracy	Precision	Recall	F1 Score
ResNet50	F: 0.7986	F: 0.6646	F: 0.3821	F: 0.4879
	H: 0.8059	H: 0.6807	H: 0.4211	H: 0.5160
VGG19	F: 0.7573	F: 0.5301	F: 0.2582	F: 0.3428
	H: 0.7638	H: 0.5530	H: 0.2871	H: 0.3730
MobileNet	F: 0.7796	F: 0.6066	F: 0.3364	F: 0.4287
	H: 0.7921	H: 0.6415	H: 0.3821	H: 0.4760



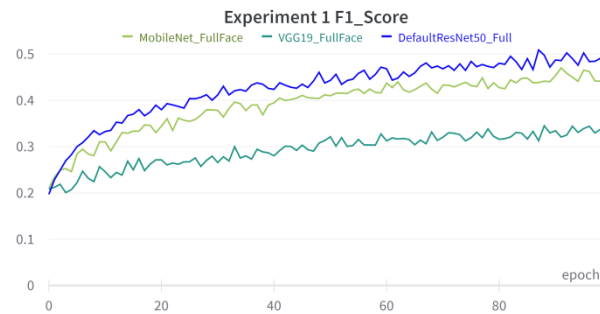
a: Training Accuracy



b: Training Precision



c: Training Recall

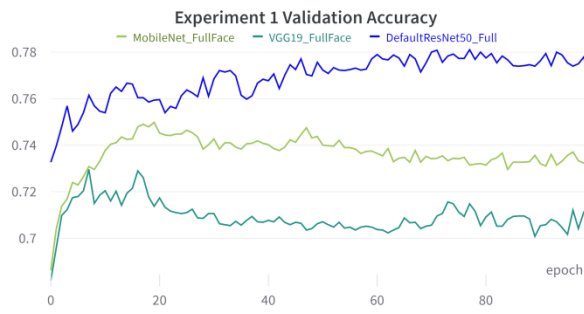


d: Training F1 Score

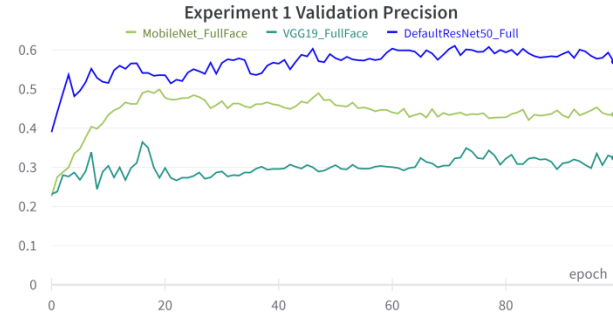
Figure 5-3: Experiment 1 Training performances.

Table 5-3: Experiment 1 Validation Results.

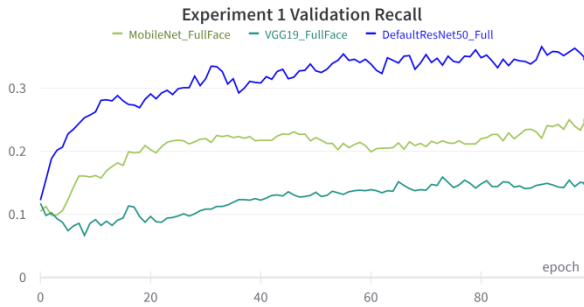
Model	Validation Accuracy	Validation Precision	Validation Recall	Validation F1 Score
ResNet50	F: 0.7713 H: 0.7385	F: 0.5699 H: 0.4671	F: 0.3467 H: 0.3258	F: 0.4250 H: 0.3806
VGG19	F: 0.7096 H: 0.7346	F: 0.3243 H: 0.4294	F: 0.1492 H: 0.1875	F: 0.211 H: 0.2578
MobileNet	F: 0.7315 H: 0.7100	F: 0.4352 H: 0.3756	F: 0.2492 H: 0.2417	F: 0.3141 H: 0.2917



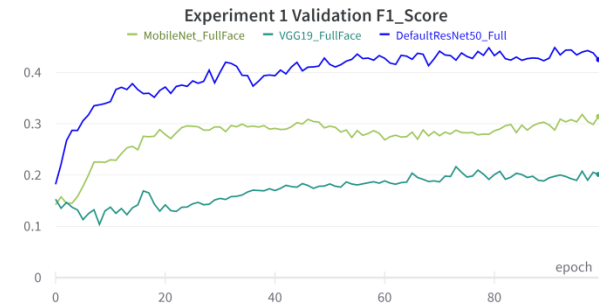
a: Validation Accuracy



b: Validation Precision



c: Validation Recall



d: Validation F1 Score

Figure 5-4: Experiment 1 Validation performances.

Table 5-4: Experiment 1 Prediction Results

Model	Data Class	Precision	Recall	F1 Score
ResNet50	Boredom	F: 0.45 H: 0.50	F: 0.61 H: 0.51	F: 0.52 H: 0.51
	Confusion	F: 0.47 H: 0.33	F: 0.37 H: 0.50	F: 0.41 H: 0.40
	Neutral	F: 0.56 H: 0.48	F: 0.46 H: 0.34	F: 0.51 H: 0.40
	Happiness	F: 0.66 H: 0.62	F: 0.69 H: 0.49	F: 0.68 H: 0.55
	Average	F: 0.5370 H: 0.4845	F: 0.5332 H: 0.4579	F: 0.5294 H: 0.4619
VGG 19	Boredom	F: 0.46 H: 0.44	F: 0.24 H: 0.30	F: 0.32 H: 0.36
	Confusion	F: 0.06 H: 0.24	F: 0.06 H: 0.27	F: 0.06 H: 0.25
	Neutral	F: 0.33 H: 0.46	F: 0.36 H: 0.50	F: 0.34 H: 0.48
	Happiness	F: 0.36 H: 0.39	F: 0.53 H: 0.43	F: 0.43 H: 0.41
	Average	F: 0.3030 H: 0.3836	F: 0.2972 H: 0.3752	F: 0.2870 H: 0.3753
MobileNet	Boredom	F: 0.33 H: 0.27	F: 0.13 H: 0.74	F: 0.19 H: 0.40
	Confusion	F: 0.18 H: 0.36	F: 0.18 H: 0.36	F: 0.18 H: 0.36
	Neutral	F: 0.33 H: 0.38	F: 0.09 H: 0.07	F: 0.14 H: 0.12
	Happiness	F: 0.25 H: 0.40	F: 0.59 H: 0.04	F: 0.35 H: 0.08
	Average	F: 0.2719 H: 0.3545	F: 0.2455 H: 0.3032	F: 0.2130 H: 0.2385

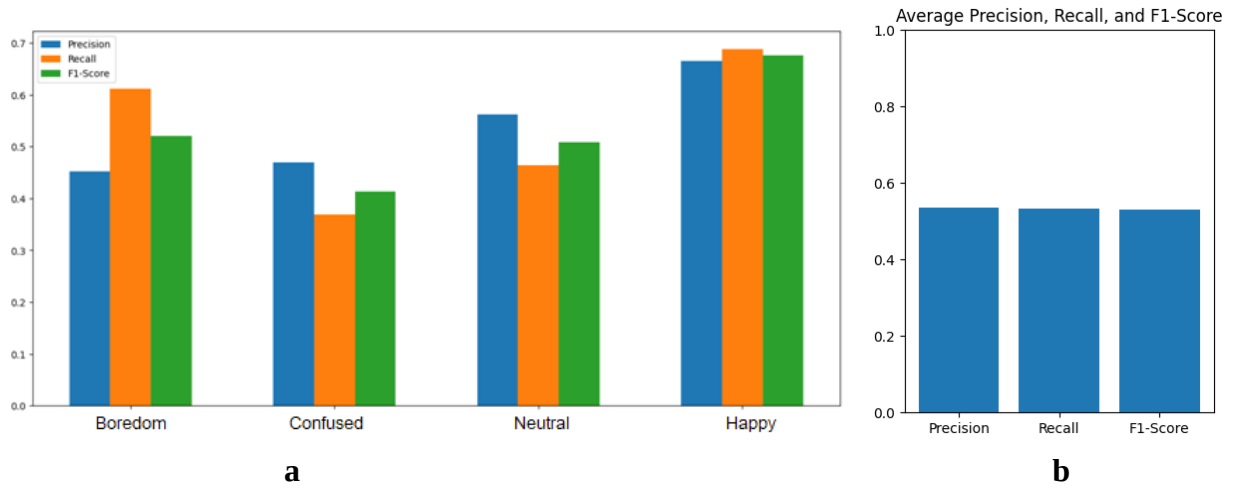


Figure 5-5: a: Default ResNet50 Prediction Results for each class (Full-face).

Figure 5-6: b: Default ResNet50 Average Prediction Results for all classes (Full-face)

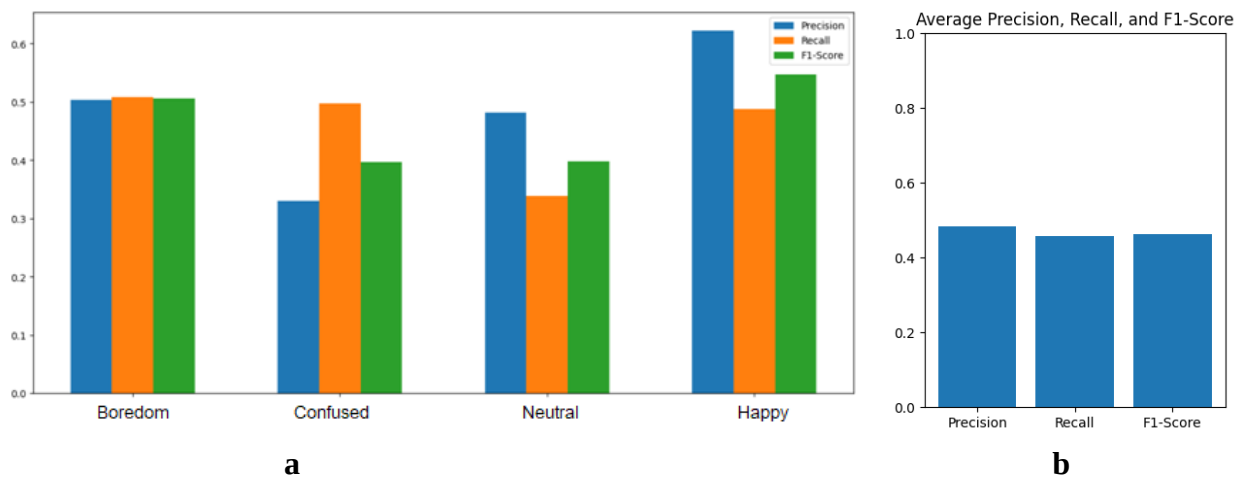


Figure 5-7: a: Default ResNet50 Prediction Results for each class (Half-face)

Figure 5-8: 3b: Default ResNet50 Average Prediction Results for all classes (Half-face)

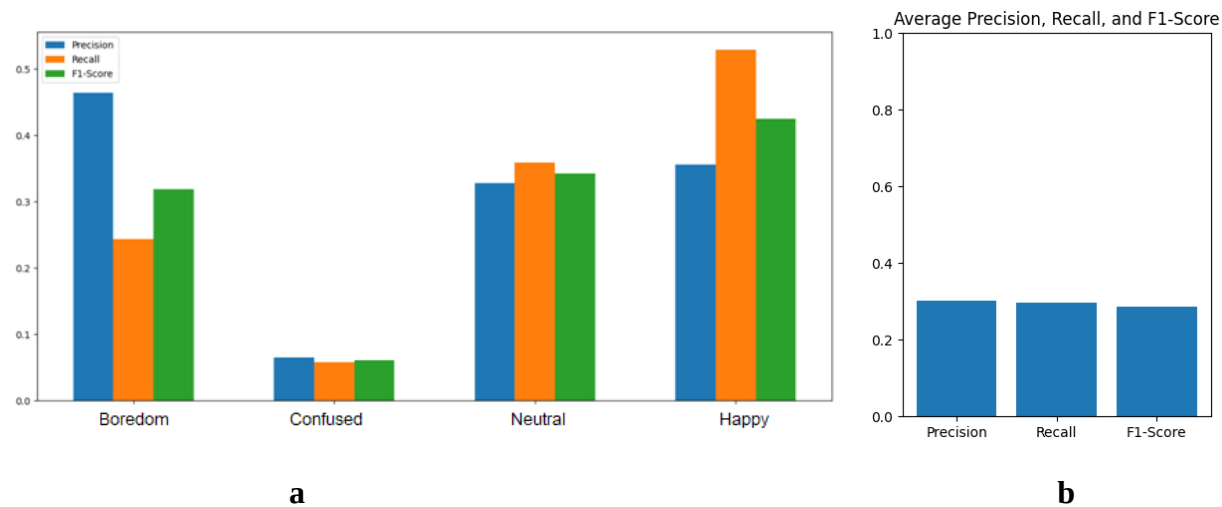


Figure 5-9: a: VGG19 Prediction Results for each class (Full-face).

Figure 5-10: b: VGG19 Average Prediction Results for all classes (Full-face)

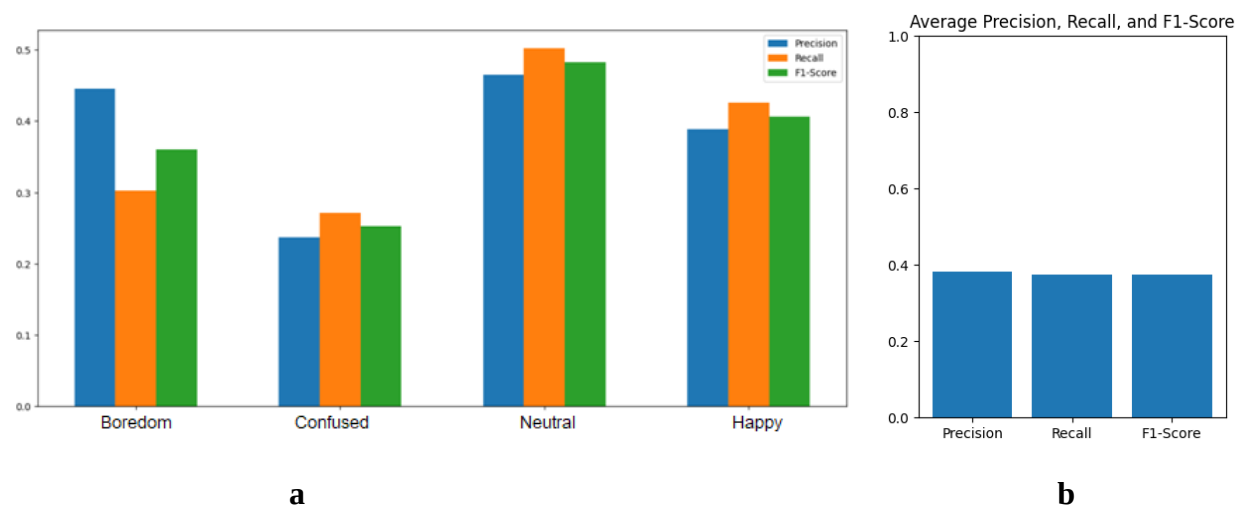


Figure 5-11: a: VGG19 Prediction Results for each class (Half-face).

Figure 5-12: b: VGG19 Average Prediction Results for all classes (Half-face)

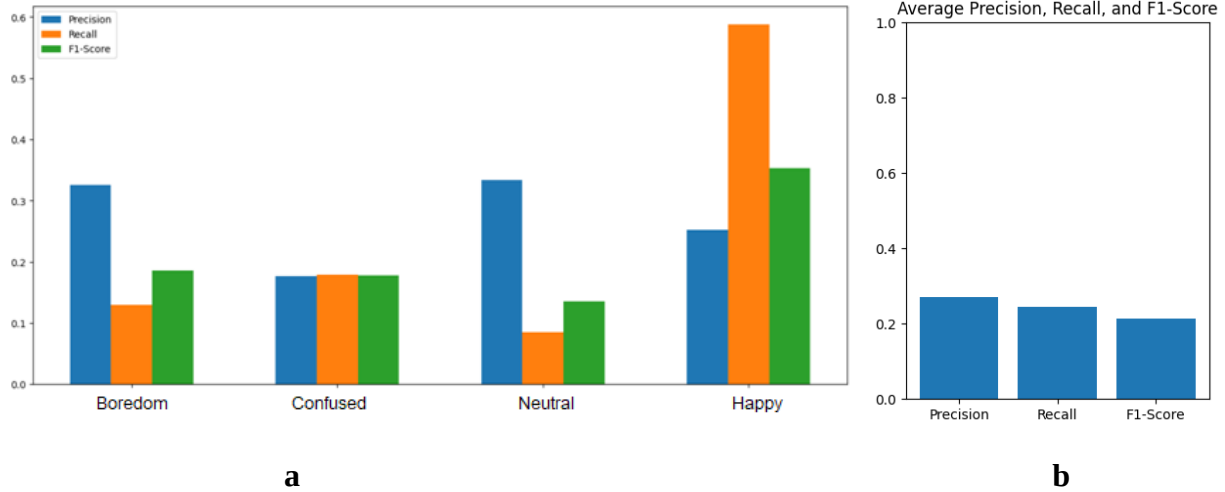


Figure 5-13: a: MobileNet Prediction Results for each class (Full-face).

Figure 5-14: b: MobileNet Average Prediction Results for all classes (Full-face)

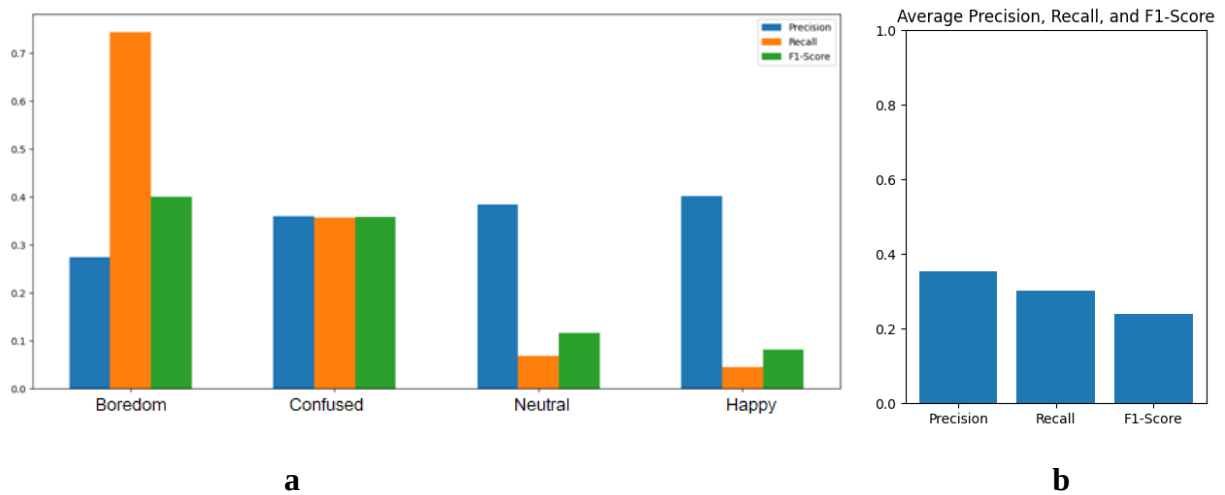


Figure 5-15: a: MobileNet Prediction Results for each class (Half-face).

Figure 5-16: b: MobileNet Average Prediction Results for all classes (Half-face)

5.2.1.3 Models Performance

A critical aspect of evaluating the efficacy of different Convolutional Neural Network models was not only their accuracy in emotion recognition but also the efficiency of their prediction processes. The effectiveness of the ResNet50, VGG19, and MobileNet models in this

this research is therefore evaluated based on their prediction latency. The following table presents a comparative overview of each model's prediction time, reflecting their capacity to process and analyze facial expressions swiftly and accurately. Each model was predicted using a custom-captured dataset. This measure directly impacts the feasibility of integrating these models into VR environments where immediate feedback and interaction are paramount.

Table 5-5. Comparison of CNN Models' Prediction Time.

Models	Prediction Time	Dataset
ResNet50	2321s (38.6 minutes)	Our specialized dataset
VGG19	2296s (38.2 minutes)	Our specialized dataset
MobileNet	652s (10.8 minutes)	Our specialized dataset

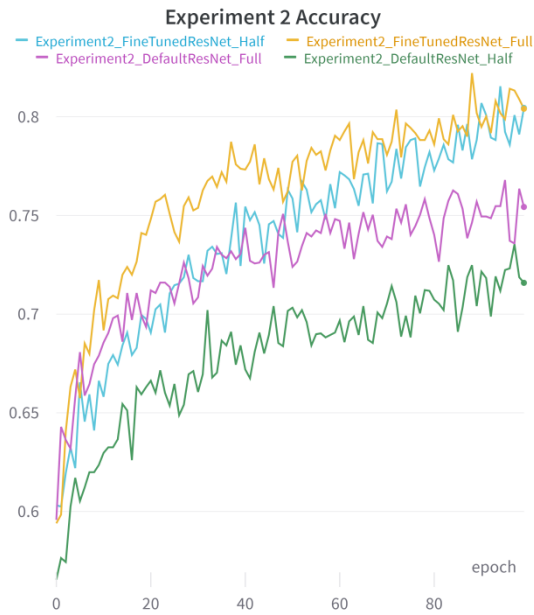
5.2.2 Experiment 2: Effectiveness When Testing on a Different Dataset

In the second experiment, we aimed to evaluate the generalizability and adaptability of our custom training approach ResNet50 model by training it on an online dataset and test its effectiveness with our own dataset, which is a different dataset that more closely resembles a VR classroom setting. We contrasted this with the performance of the default ResNet50 model. This was conducted to test the robustness of our customized training approach model in working with new, unseen data. All the Tables and Figures in this segment of our experiment 2 show the results.

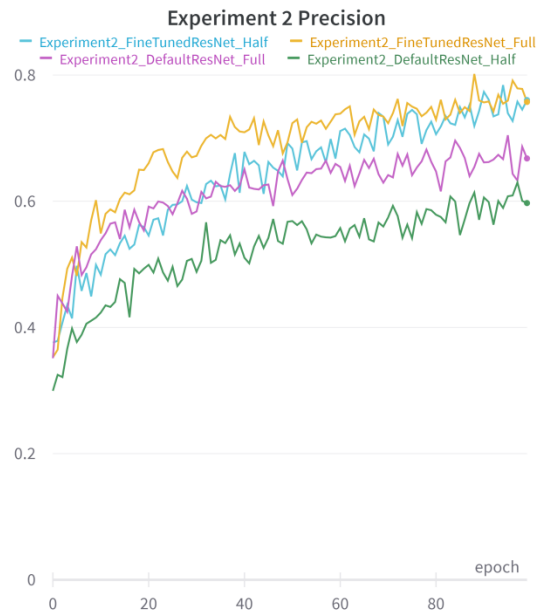
We trained both the default ResNet50 and our customized version on the AffectNet dataset. Then we used these trained models to make predictions on our dataset. The performance metrics shows that our customized ResNet50 model performed better than the default ResNet50 model in making accurate predictions on our specialized dataset. This highlights the versatility and generalizability of our customized training approach model when faced with new data.

Table 5-6: Experiment 2 Training Results. (Full Face = F, Half Face = H)

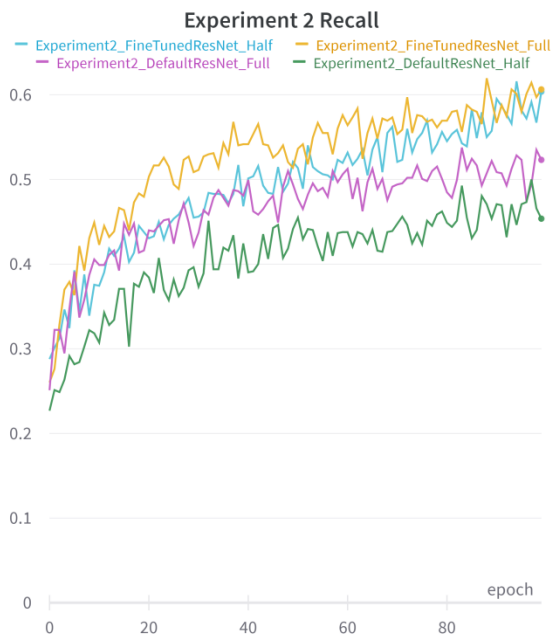
Model	Accuracy	Precision	Recall	F1 Score
Default ResNet50	F: 0.7543	F: 0.6678	F: 0.5231	F: 0.5823
	H: 0.7159	H: 0.5971	H: 0.4537	H: 0.5117
Custom Approach ResNet50	F: 0.8041	F: 0.7574	F: 0.6063	F: 0.6739
	H: 0.8045	H: 0.7604	H: 0.6037	H: 0.6716



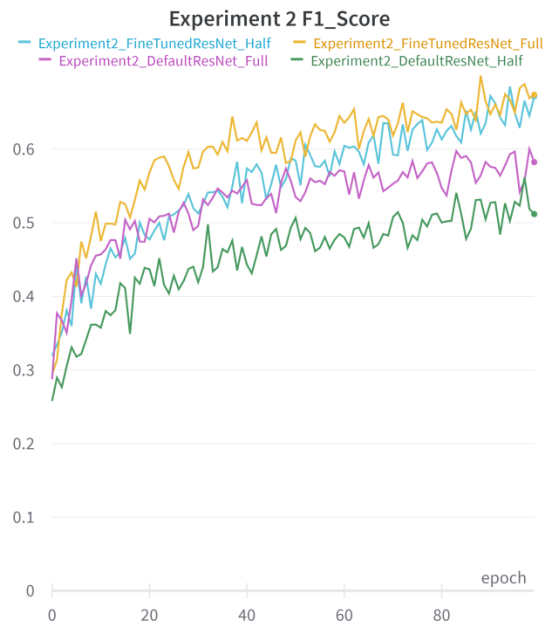
a: Training Accuracy



b: Training Precision



c: Training Recall

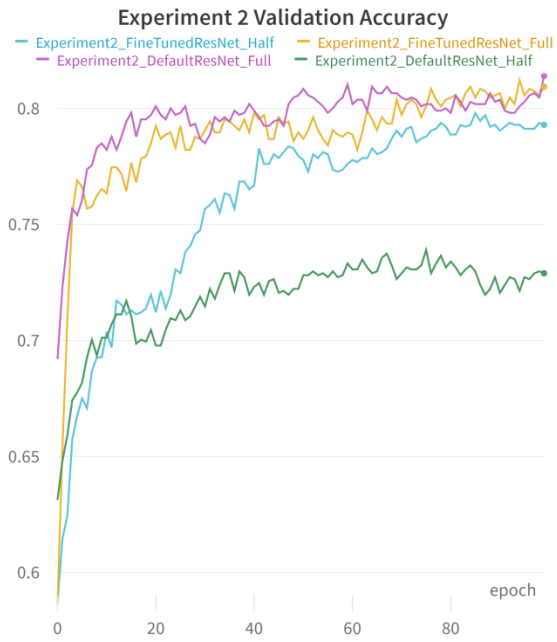


d: Training F1 Score

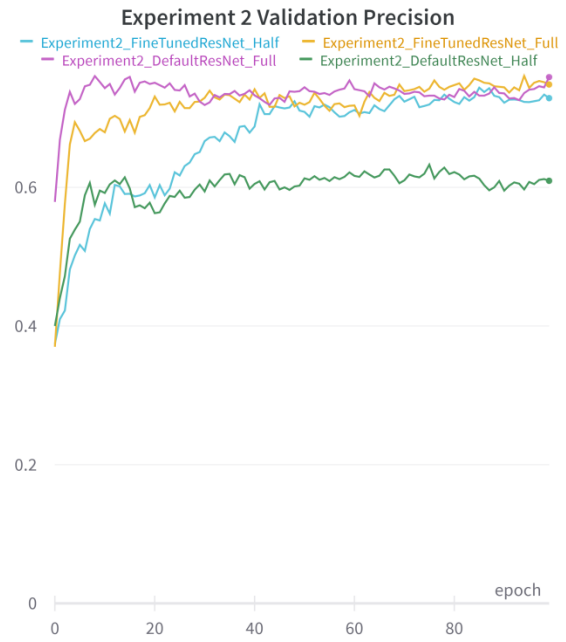
Figure 5-17: Experiment 2 training performances. Both Full-face and Half-face.

Table 5-7: Experiment 2 Validation Results.

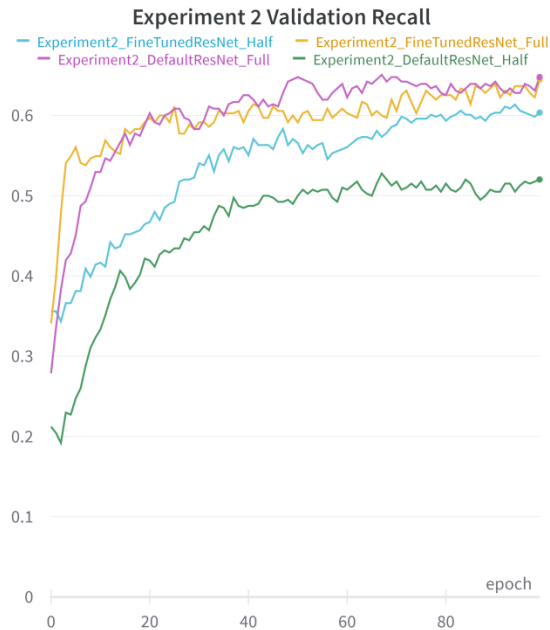
Model	Val Accuracy	Val Precision	Val Recall	Val F1 Score
Default ResNet50	F: 0.8141	F: 0.7591	F: 0.6479	F: 0.6975
	H: 0.7290	H: 0.6095	H: 0.5202	H: 0.5567
Customized ResNet50	F: 0.8094	F: 0.7484	F: 0.6451	F: 0.6804
	H: 0.7929	H: 0.7287	H: 0.6035	H: 0.6578



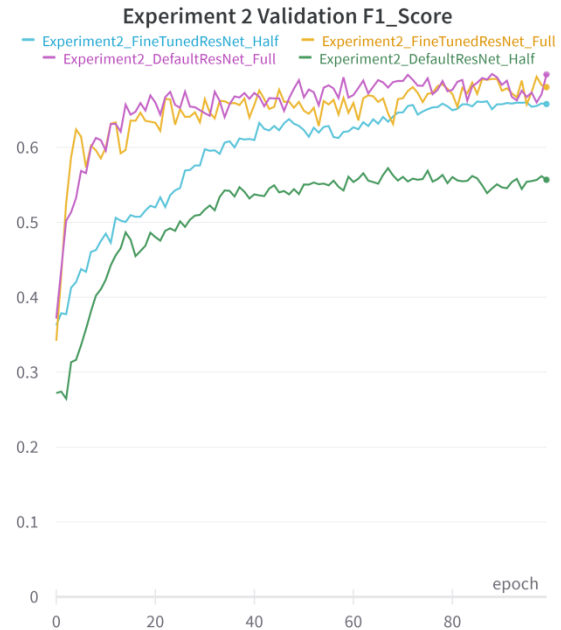
a: Validation Accuracy



b: Validation Precision



c: Validation Recall



d: Validation F1 Score

Figure 5-18: Experiment 2 Validation performances. Both Full-face and Half-face.

Table 5-8: Experiment 2 Prediction Results

Model	Data Class	Precision	Recall	F1 Score
Default ResNet50	Boredom	F: 0.43 H: 0.44	F: 0.46 H: 0.58	F: 0.44 H: 0.50
	Neutral	F: 0.39 H: 0.47	F: 0.51 H: 0.59	F: 0.44 H: 0.52
	Happiness	F: 0.69 H: 0.88	F: 0.43 H: 0.36	F: 0.53 H: 0.51
	Average	F: 0.5033 H: 0.5948	F: 0.4666 H: 0.511	F: 0.47 H: 0.5113
Custom Approach ResNet50	Boredom	F: 0.61 H: 0.49	F: 0.46 H: 0.61	F: 0.52 H: 0.54
	Neutral	F: 0.66 H: 0.46	F: 0.72 H: 0.45	F: 0.69 H: 0.46
	Happiness	F: 0.76 H: 0.76	F: 0.82 H: 0.60	F: 0.79 H: 0.67
	Average	F: 0.6766 H: 0.57	F: 0.6666 H: 0.5533	F: 0.6666 H: 0.5566

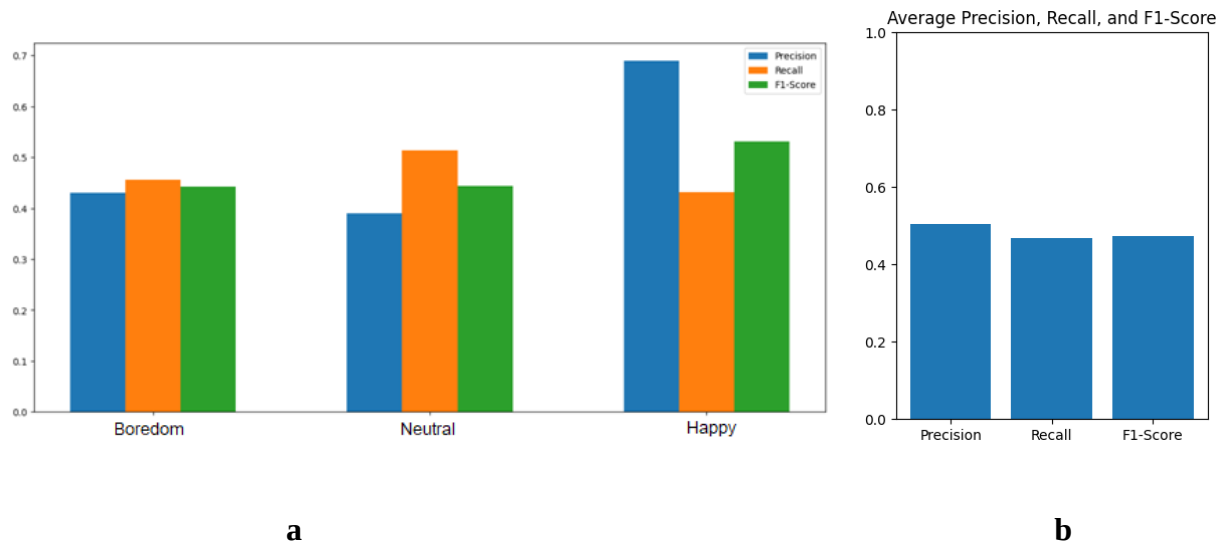


Figure 5-19: a: Default ResNet50 Prediction Results for each class (Full-face).

Figure 5-20: b: Default ResNet50 Average Prediction Results for all classes (Full-face)

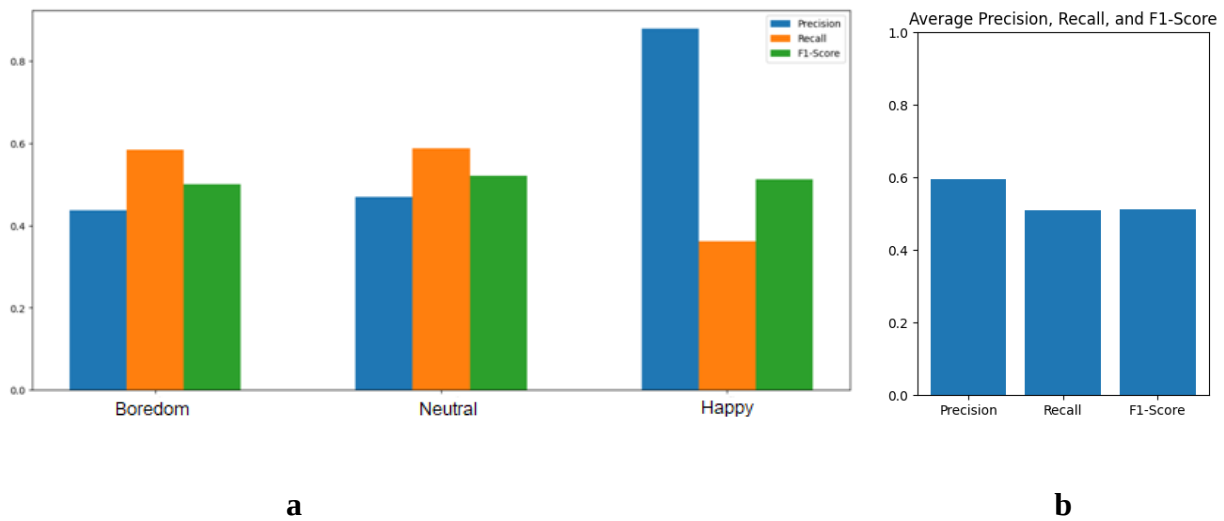
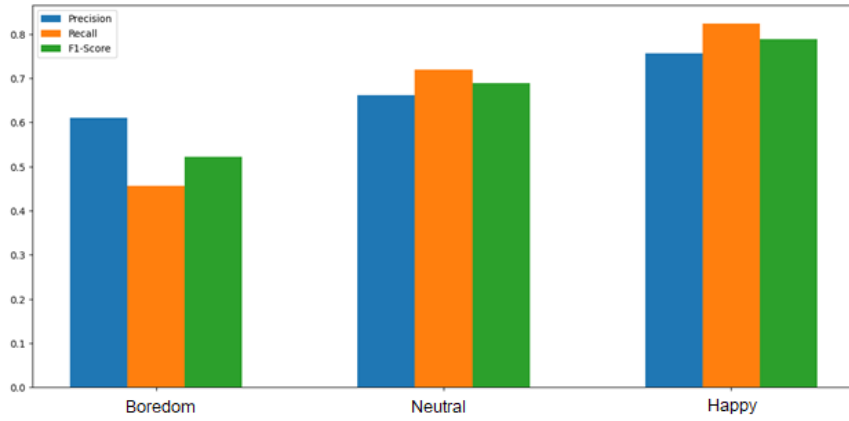
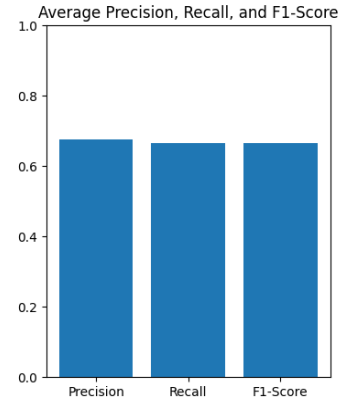


Figure 5-21: a: Default ResNet50 Prediction Results for each class (Half-face).

Figure 5-22: b: Default ResNet50 Average Prediction Results for all classes (Half-face)



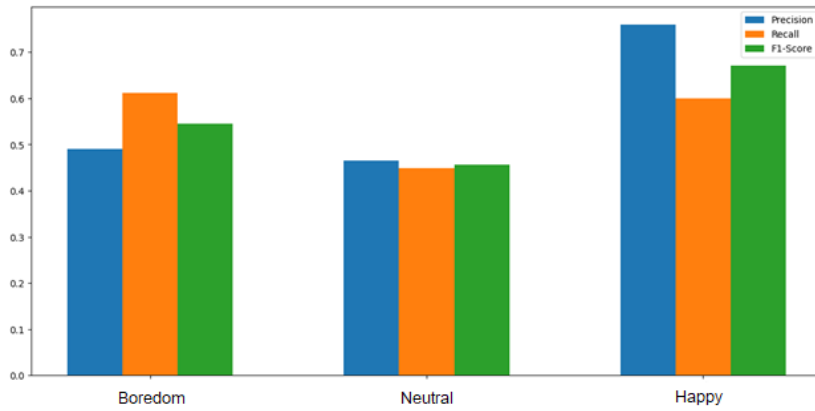
a



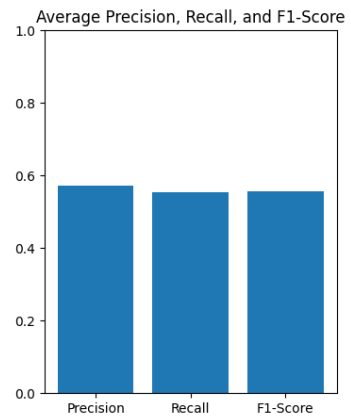
b

Figure 5-23: a: Custom Approach ResNet50 Prediction Results for each class (Full-face).

Figure 5-24: b: Custom Approach ResNet50 Average Prediction Results for all classes (Full-face)



a



b

Figure 5-25: a: Custom Approach ResNet50 Prediction Results for each class (Half-face).

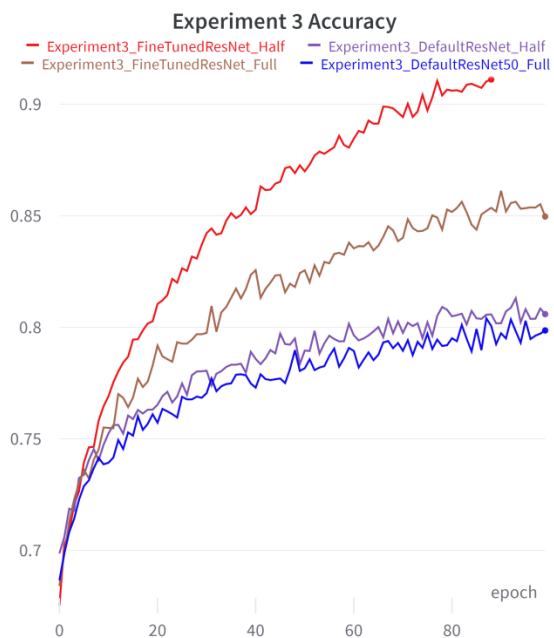
Figure 5-26: b: Custom Approach ResNet50 Average Prediction Results for all classes (Half-face)

5.2.3 Experiment 3: Default ResNet50 vs Custom Approach ResNet50

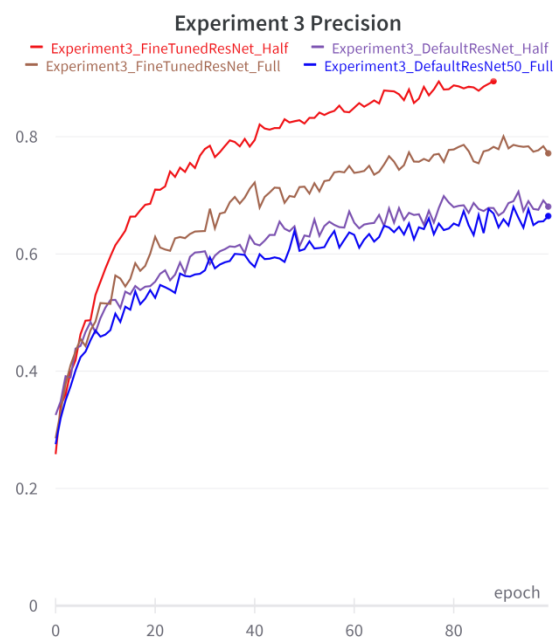
Our third experiment aimed to evaluate the performance of our custom approach model using only our specialized dataset. This was like Experiment 2, but in this case, we partitioned our specialized dataset into a training set, a validation set, and prediction set, focusing on both full-face and occluded half-face images. The models were then trained, validated, and predicted using these sets. All the Tables and Figures in this segment of our experiment 3 show the results. Our customized training approach model again outperformed the default ResNet50 model, irrespective of the full face or half face context.

Table 5-9: Experiment 3 Training Results. (Full Face = F, Half Face = H)

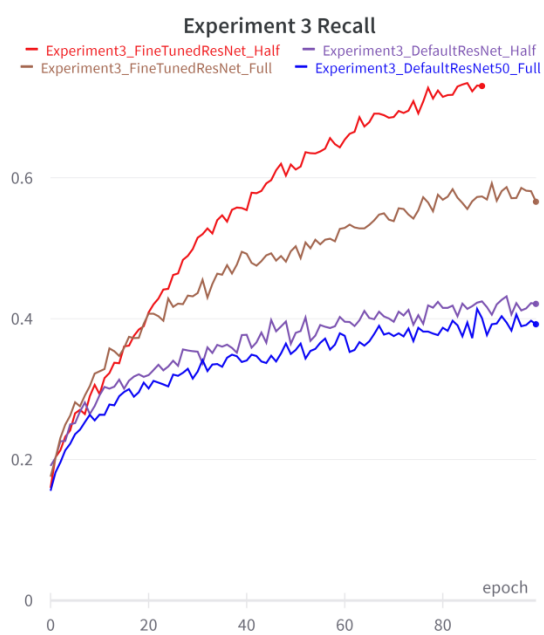
Model	Accuracy	Precision	Recall	F1 Score
Default ResNet50	F: 0.7986	F: 0.6646	F: 0.3821	F: 0.4879
	H: 0.8059	H: 0.6807	H: 0.4211	H: 0.5160
Custom Approach ResNet50	F: 0.8496	F: 0.7717	F: 0.5661	F: 0.6502
	H: 0.9111	H: 0.8946	H: 0.7304	H: 0.8033



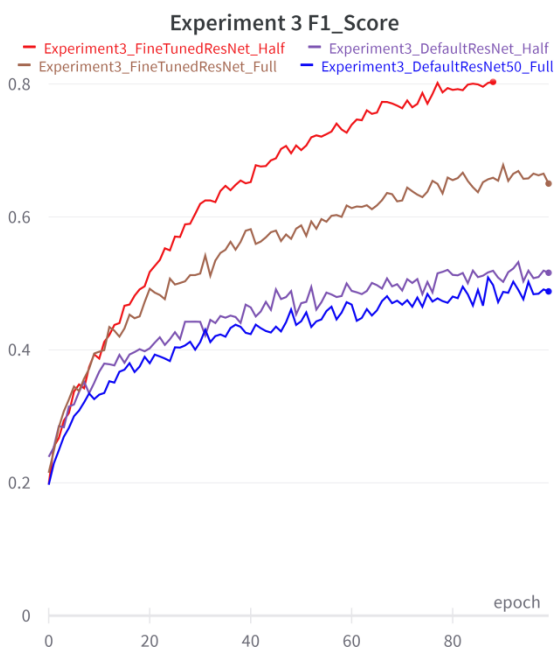
c: Training Accuracy



d: Training Precision



c: Training Recall

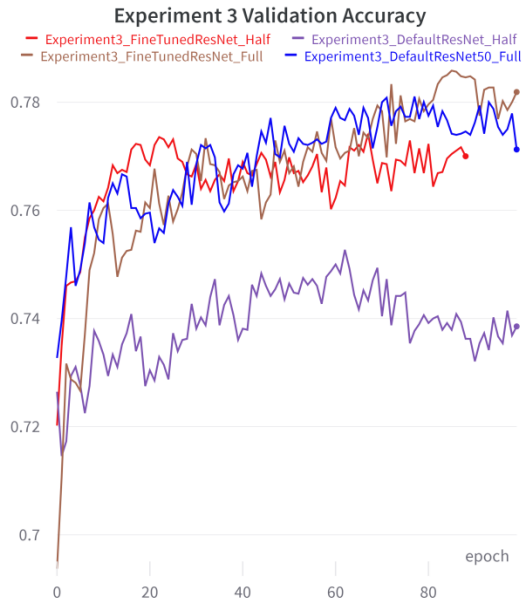


d: Training F1 Score

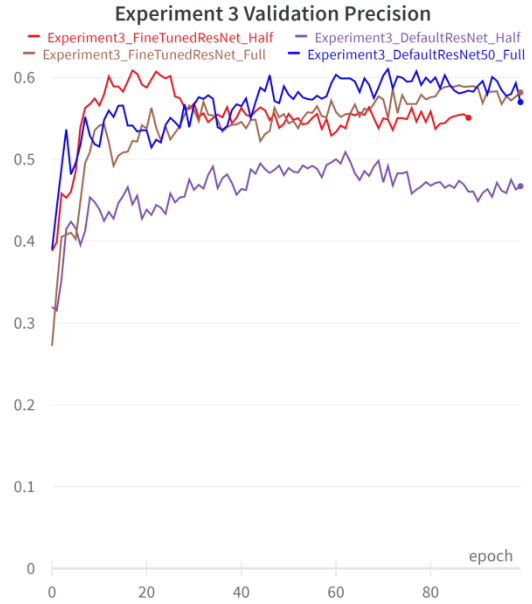
Figure 5-27: Experiment 3 training performances. Both Full-face and Half-face.

Table 5-10: Experiment 3 Validation Results.

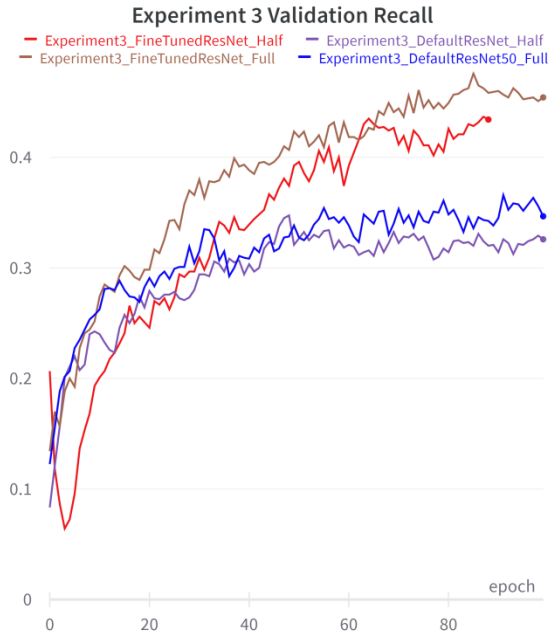
Model	Val Accuracy	Val Precision	Val Recall	Val F1 Score
Default ResNet50	F: 0.7713	F: 0.5699	F: 0.3467	F: 0.4250
	H: 0.7385	H: 0.4671	H: 0.3258	H: 0.3806
Custom Approach ResNet50	F: 0.7819	F: 0.5816	F: 0.4542	F: 0.5075
	H: 0.7700	H: 0.5507	H: 0.4342	H: 0.4846



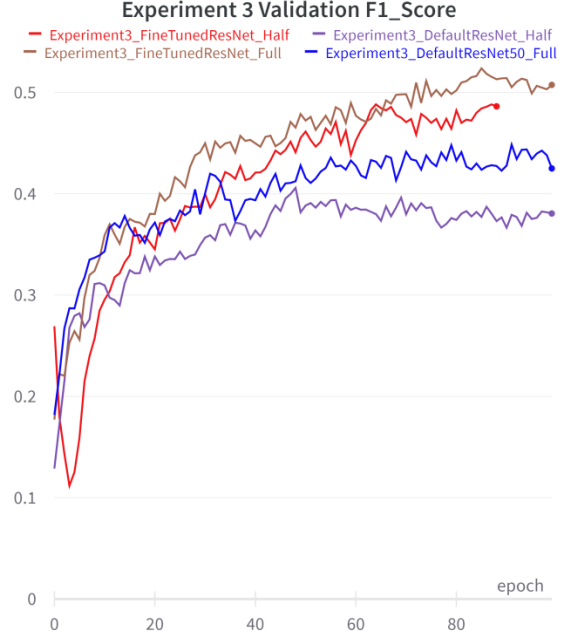
a: Validation Accuracy



b: Validation Precision



c: Validation Recall



d: Validation F1 Score

Figure 5-28: Experiment 3 Validation performances. Both Full-face and Half-face.

Table 5-11: Experiment 3 Prediction Results.

Model	Data Class	Precision	Recall	F1 Score
Default ResNet50	Boredom	F: 0.45 H: 0.50	F: 0.61 H: 0.51	F: 0.52 H: 0.51
	Confusion	F: 0.47 H: 0.33	F: 0.37 H: 0.50	F: 0.41 H: 0.40
	Neutral	F: 0.56 H: 0.48	F: 0.46 H: 0.34	F: 0.51 H: 0.40
	Happiness	F: 0.66 H: 0.62	F: 0.69 H: 0.49	F: 0.68 H: 0.55
	Average	F: 0.5370 H: 0.4845	F: 0.5332 H: 0.4579	F: 0.5294 H: 0.4619
	Boredom	F: 0.71 H: 0.58	F: 0.51 H: 0.61	F: 0.59 H: 0.60
	Confusion	F: 0.74	F: 0.70	F: 0.72

Custom Approach ResNet50		H: 0.45	H: 0.57	H: 0.50
	Neutral	F: 0.63 H: 0.75	F: 0.82 H: 0.64	F: 0.71 H: 0.69
	Happiness	F: 0.80 H: 0.77	F: 0.85 H: 0.65	F: 0.82 H: 0.70
	Average	F: 0.7224 H: 0.6379	F: 0.7177 H: 0.6145	F: 0.7125 H: 0.6217

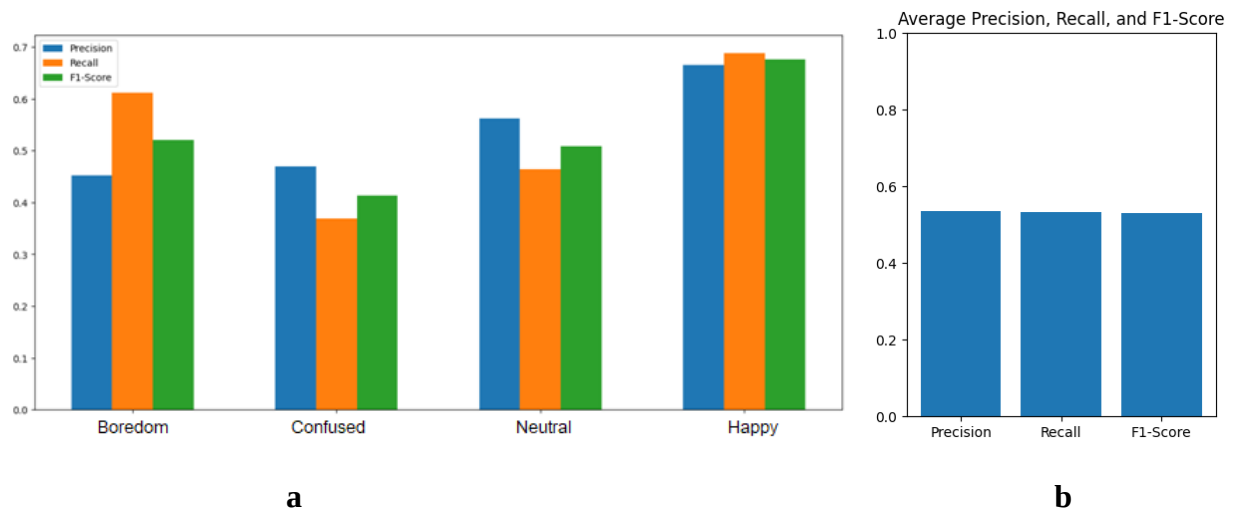


Figure 5-29: a: Default ResNet50 Prediction Results for each class (Full-face).

Figure 5-30: b: Default ResNet50 Average Prediction Results for all classes (Full-face)

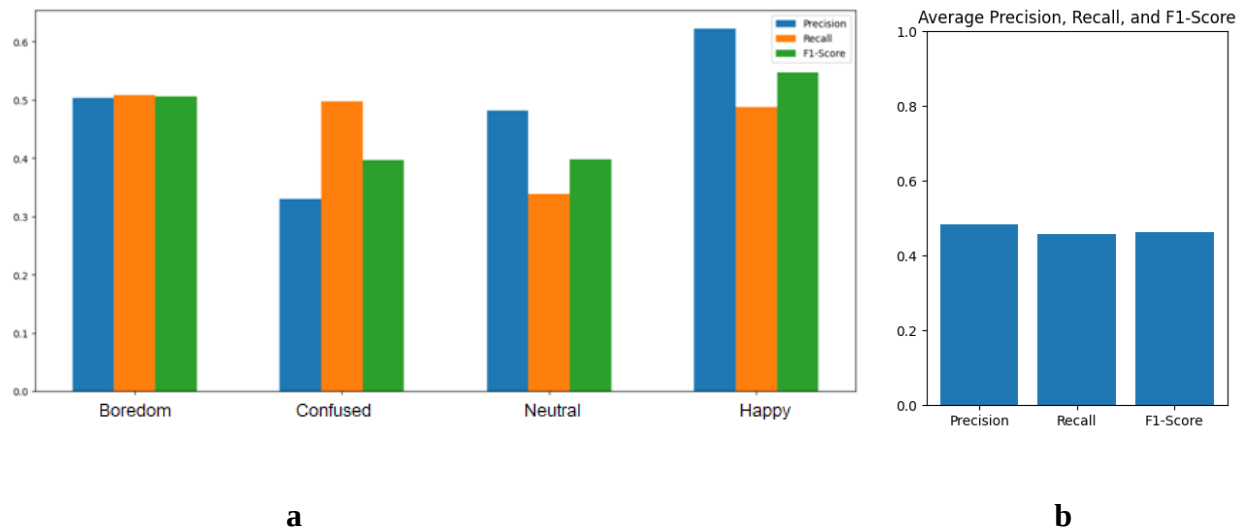


Figure 5-31: a: Default ResNet50 Prediction Results for each class (Half-face).

Figure 5-32: b: Default ResNet50 Average Prediction Results for all classes (Half-face)

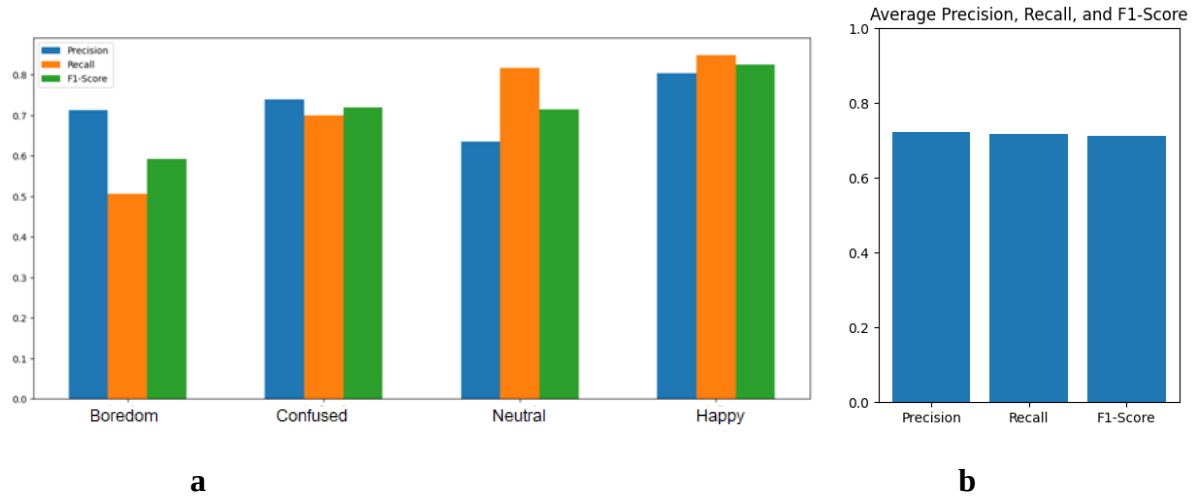


Figure 5-33: a: Custom Approach ResNet50 Prediction Results for each class (Full-face).

Figure 5-34: b: Custom Approach ResNet50 Average Prediction Results for all classes (Full-face)

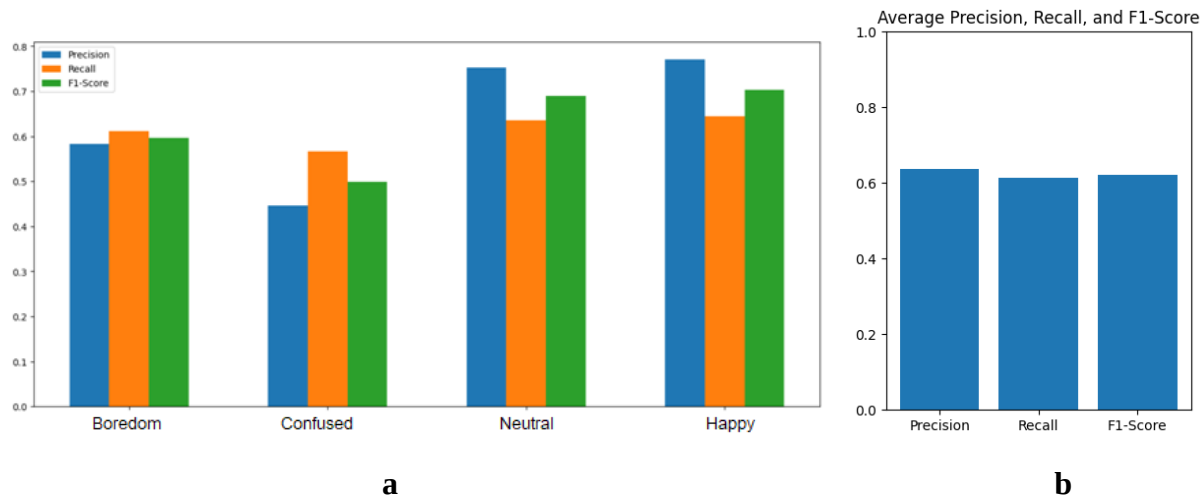


Figure 5-35: a: Custom Approach ResNet50 Prediction Results for each class (Half-face).

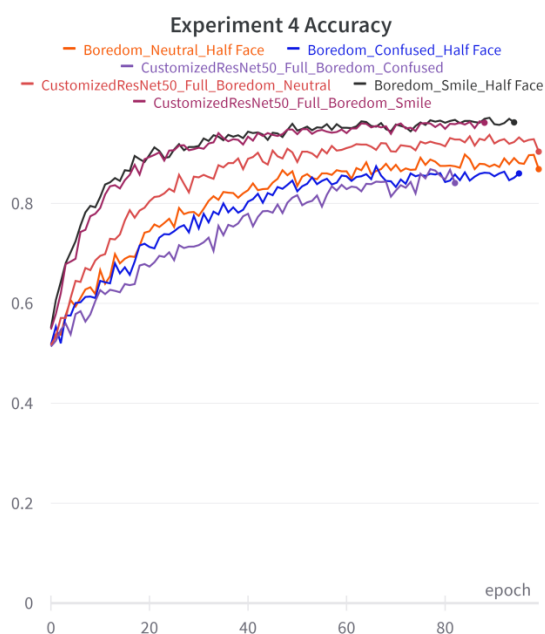
Figure 5-36: b: Custom Approach ResNet50 Average Prediction Results for all classes (Half-face)

5.2.4 Experiment 4: Binary Classification Experiments Involving Boredom

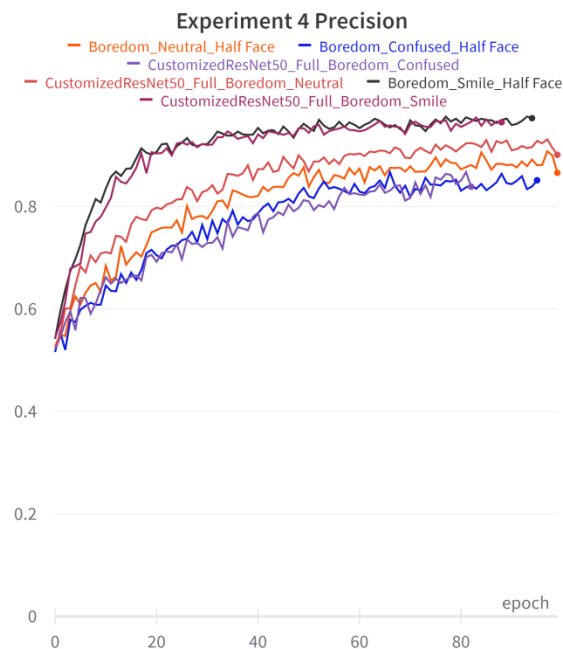
The fourth experiment aimed to test the discriminatory power of our custom approach model, specifically in differentiating “Boredom” from other expressions like “Neutral,” “Happiness” and “Confusion.” This is of particular interest as the expression for “Boredom” often closely resembles “Neutral,” making it a challenging task. We configured the models to perform binary classifications. We set up three separate scenarios: Boredom vs. Neutral, Boredom vs. Happiness, and Boredom vs. Confusion. The experiment was conducted using both full-face and occluded half-face datasets. All the Tables and Figures in this segment of our experiment 4 show the results. The result of this experiment provided intriguing insights. Not only did our custom approach model manage to effectively differentiate “Boredom” from other expressions, but it also achieved a reasonable level of accuracy across both full-face and occluded half-face conditions.

Table 5-12: Experiment 4 Training Results. (Full Face = F, Half Face = H)

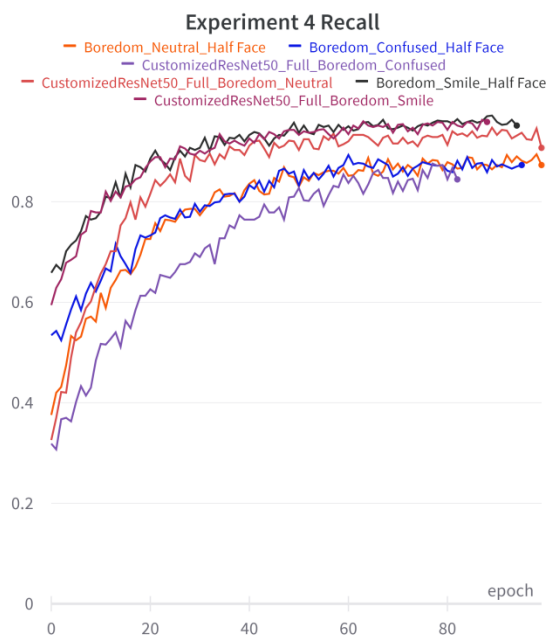
Data Classes	Accuracy	Precision	Recall	F1 Score
Boredom vs Neutral	F: 0.9036	F: 0.9007	F: 0.9071	F: 0.9012
	H: 0.8654	H: 0.8654	H: 0.8729	H: 0.8646
Boredom vs Smile	F: 0.9614	F: 0.9641	F: 0.9586	F: 0.9577
	H: 0.9621	H: 0.9723	H: 0.9514	H: 0.9603
Boredom vs Confusion	F: 0.8407	F: 0.8383	F: 0.8443	F: 0.8368
	H: 0.8600	H: 0.8510	H: 0.8729	H: 0.8598



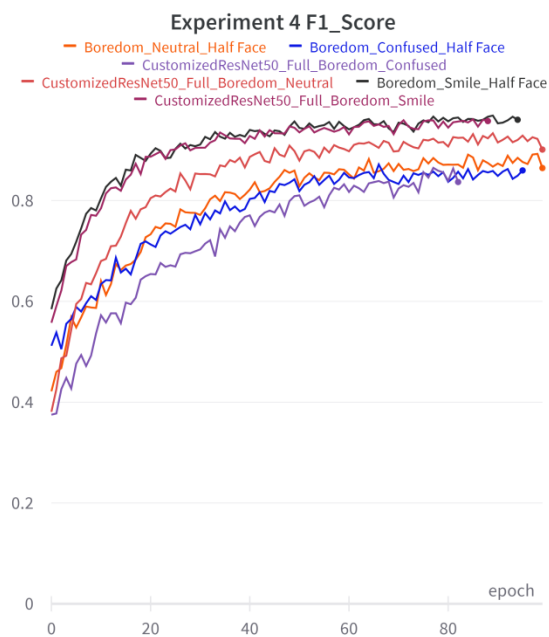
a: Training Accuracy



b: Training Precision



c: Training Recall



d: Training F1 Score

Figure 5-37: Experiment 4 training performances. Both Full-face and Half-face.

Table 5-13: Experiment 4 Validation Results. (Full Face = F, Half Face = H)

Data Classes	Val Accuracy	Val Precision	Val Recall	Val F1 Score
Boredom vs Neutral	F: 0.6217 H: 0.6300	F: 0.6096 H: 0.6585	F: 0.6767 H: 0.5400	F: 0.6358 H: 0.5899
Boredom vs Smile	F: 0.7917 H: 0.8267	F: 0.7814 H: 0.8630	F: 0.8100 H: 0.7767	F: 0.7826 H: 0.8076
Boredom vs Confusion	F: 0.6117 H: 0.6133	F: 0.6128 H: 0.6453	F: 0.6067 H: 0.5033	F: 0.5902 H: 0.5451

Table 5-14: Experiment 4 Prediction Results

Data Class	Precision	Recall	F1 Score
Boredom	F: 0.63 H: 0.67	F: 0.76 H: 0.57	F: 0.69 H: 0.61
Neutral	F: 0.69 H: 0.62	F: 0.55 H: 0.72	F: 0.61 H: 0.67
Average	F: 0.6596 H: 0.6442	F: 0.653 H: 0.641	F: 0.6493 H: 0.6389
Boredom	F: 0.80 H: 0.83	F: 0.90 H: 0.77	F: 0.85 H: 0.80
Smile	F: 0.89 H: 0.79	F: 0.78 H: 0.85	F: 0.83 H: 0.82
Average	F: 0.8432 H: 0.8102	F: 0.8380 H: 0.8085	F: 0.8373 H: 0.8082
Boredom	F: 0.65 H: 0.61	F: 0.63 H: 0.75	F: 0.64 H: 0.67
Confusion	F: 0.64 H: 0.67	F: 0.66 H: 0.51	F: 0.65 H: 0.58
Average	F: 0.6471 H: 0.6373	F: 0.647 H: 0.63	F: 0.6468 H: 0.6249

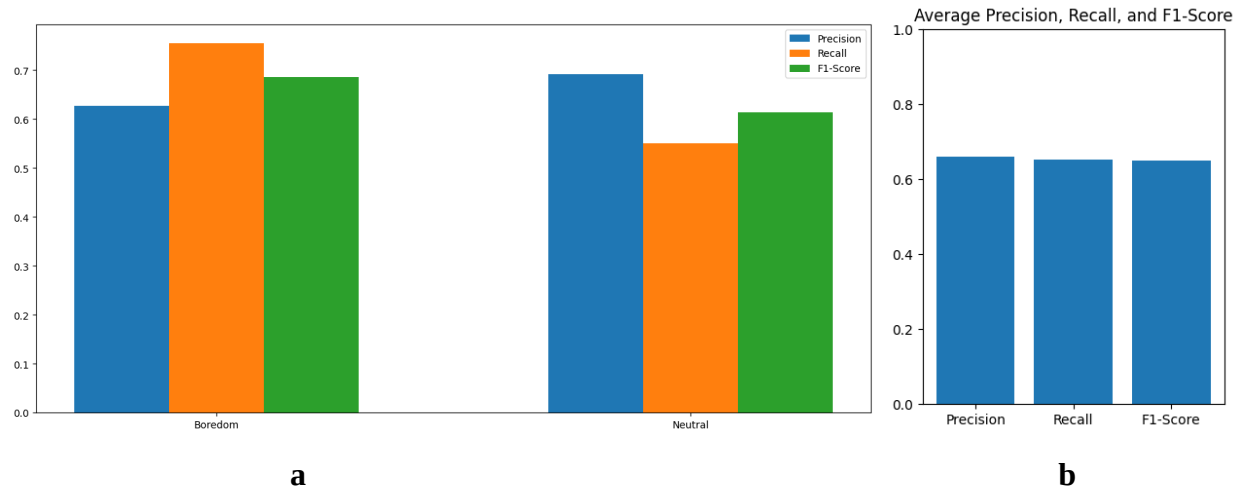


Figure 5-38: a: Custom Approach ResNet50 Prediction Results for Boredom vs Neutral (Full-face).

Figure 5-39: b: Custom Approach ResNet50 Average Prediction Results for Boredom vs Neutral (Full-face)

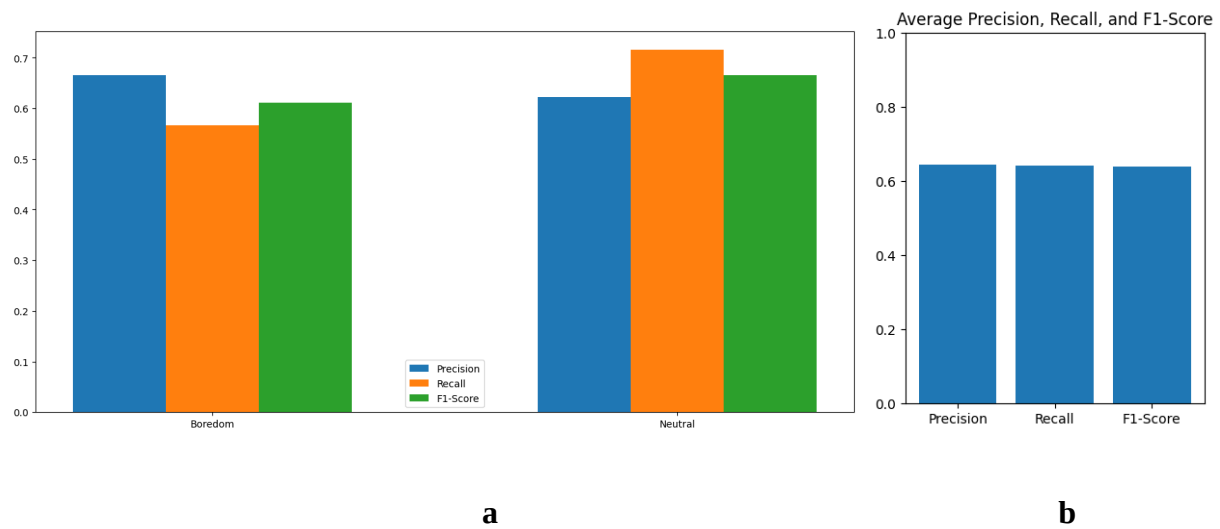
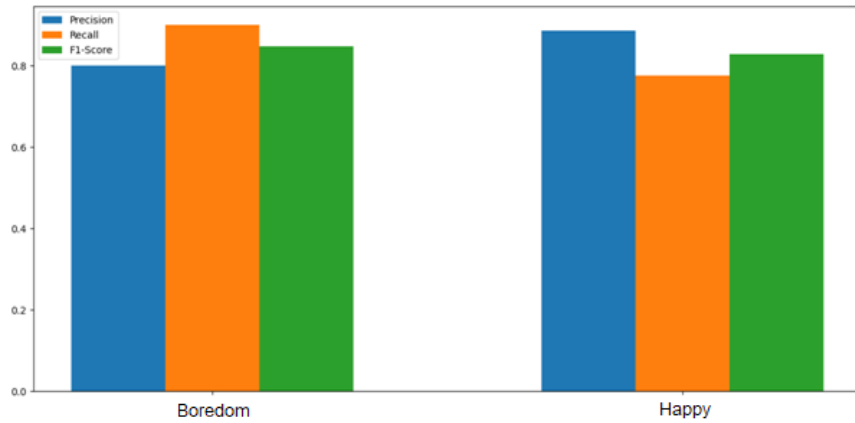
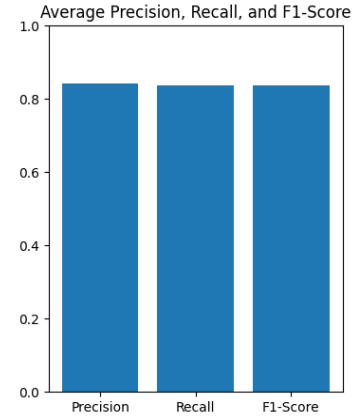


Figure 5-40: a: Custom Approach ResNet50 Prediction Results for Boredom vs Neutral (Half-face).

Figure 5-41: b: Custom Approach ResNet50 Average Prediction Results for Boredom vs Neutral (Half-face)



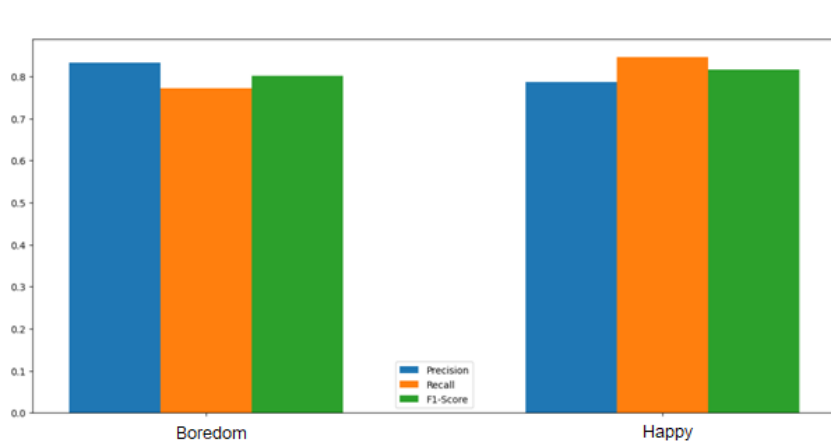
a



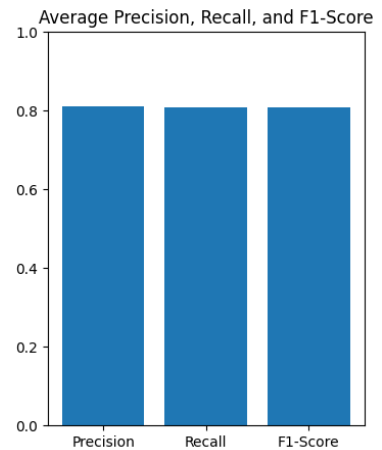
b

Figure 5-42: a: Custom Approach ResNet50 Prediction Results. Boredom vs. Happiness (Full-face).

Figure 5-43: b: Custom Approach ResNet50 Average Prediction Results for Boredom vs. Happiness (Full-face)



a



b

Figure 5-44: a: Custom Approach ResNet50 Prediction Results. Boredom vs. Happiness (Half-face).

Figure 5-45: b: Custom Approach ResNet50 Average Prediction Results for Boredom vs. Happiness (Half-face)

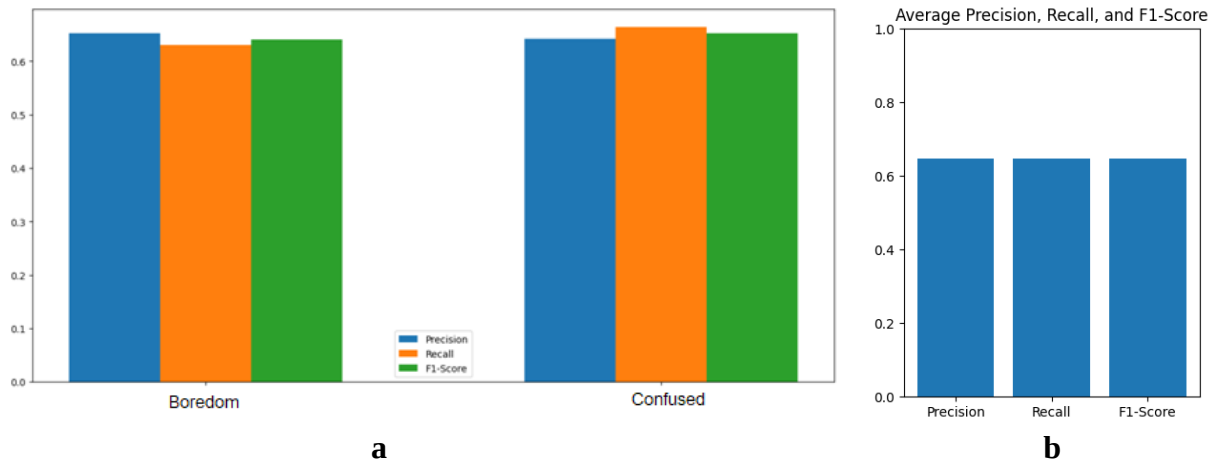


Figure 5-46: a: Custom Approach ResNet50 Prediction Results. Boredom vs. Confusion (Full-face).

Figure 5-47: b: Custom Approach ResNet50 Average Prediction Results for Boredom vs. Confusion (Full-face)

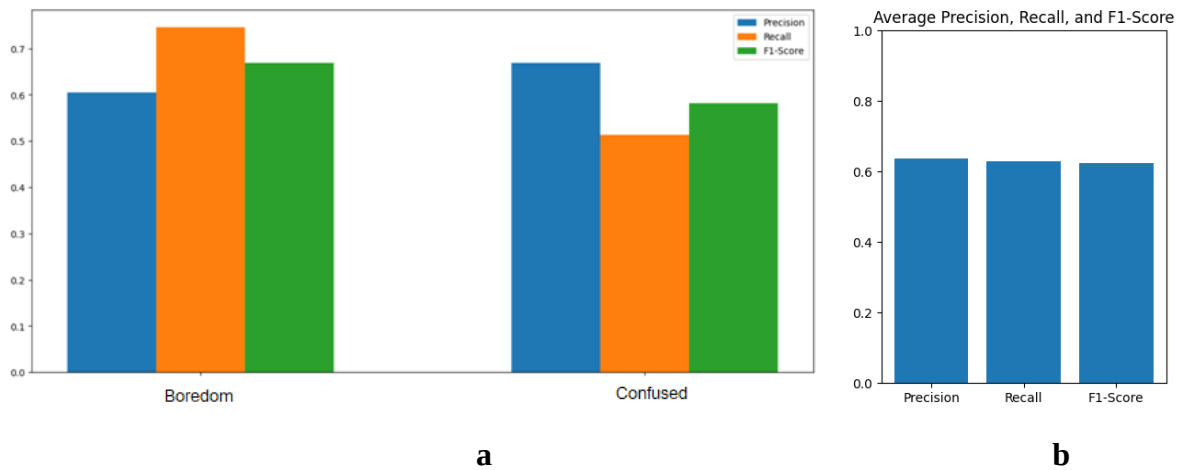


Figure 5-48: a: Custom Approach ResNet50 Prediction Results. Boredom vs. Confusion (Half-face).

Figure 5-49: b: Custom Approach ResNet50 Average Prediction Results for Boredom vs. Confusion (Half-face)

5.3 Responses to Research Questions

The importance of any research lies in its ability to systematically address and provide insights into the questions that form its foundation. In this section, we delve into an analysis and detailed responses to the primary, subsidiary, technical and implementation research questions that have guided this study. Each response is based on the findings and methodologies discussed in previous sections, particularly the experimental results and discussions. The goal is to present clear and conclusive answers that not only showcase the depth of our research but also contribute to a broader understanding of emotion recognition technologies in VR classroom settings. This section acts as a link between the framework of our research and its practical implications, ensuring a coherent narrative flow, from identifying problems to exploring potential solutions.

5.3.1 Primary Research Question

(a) ResNet50 vs. Other CNN Models: This thesis provides a detailed comparative analysis of ResNet50 with VGG19 and MobileNet, focusing on their performance in facial recognition in VR classroom settings. From our experiments (Section 5.2.1), ResNet50 exhibited superior performance in terms of accuracy and generalization when applied to both full-face and VR-occluded face datasets. This outcome highlights that while VGG19 and MobileNet have their merits, ResNet50's architecture, particularly its depth and residual connections, makes it more suitable for the complexities involved in VR-based emotion recognition.

(b) Custom Approach CNN Model Adaptation from ResNet50: Our work also explored the effectiveness of a customized training approach CNN model adapted from the ResNet50 architecture in recognizing and differentiating subtle facial expressions, such as

neutrality, boredom, happiness, and confusion, in VR classroom settings. This effectiveness is evident in (Sections 5.2.2 and 5.2.3), where our model outperformed the default ResNet50 model in accuracy and adaptability in both full-face and occluded scenarios. In Experiment 4, we also demonstrate how we were able to test our customized training approach model alongside boredom and other expressions like neutrality, happiness, and confusion, yielding decent results. Our approach to modifying the ResNet50 model, specifically targeting the lower half of the face to address occlusion challenges, appears to be a key factor in its success.

5.3.2 Subsidiary Questions:

(a) Limitations of Emotion Recognition Technologies in VR: The limitations of current emotion recognition technologies in VR environments in this thesis revolve around occlusion challenges and data diversity, including the absence of boredom facial expressions, which were missing in previous studies. Boredom is one of the essential facial expressions in educational settings (Section 6.2). Our customized approach CNN model addresses these challenges by focusing on the lower half of the face, which remains visible even with a VR headset occlusion.

(b) Impact of Occlusion on Facial Expression Recognition: Occlusion caused by VR headsets significantly affects the accuracy of facial expression recognition. This thesis incorporated techniques that improved recognition performance for the visible portions of the face, such as sourcing a public dataset called AffectNet and capturing a dataset by our research team to mimic a VR-based classroom setting. Therefore, we focused on the lower half of the face and employed data augmentation strategies to simulate various real-world conditions.

(c) Comparative Performance of default ResNet50 vs. Custom Approach ResNet50:

The comparative performance analysis, especially Experiment 3 (Section 5.2.3), between the default ResNet50 architecture and the customized approach model revealed that the latter showed improved recognition capabilities for facial expressions. This was evident in both full facial datasets and datasets partially occluded by VR headsets, underlining the effectiveness of the model's customization.

(d) Role of Data Augmentation in Enhancing Model Robustness: Data augmentation techniques were employed to enhance the robustness of the facial recognition model, particularly in handling the diversity of facial expressions in VR classroom scenarios. Techniques like random horizontal flips and rotations introduced variability to the training data, aiding the model in simulating variability that might encounter, thereby improving its ability to generalize across different facial expressions (Section 4.4.3).

5.3.3 Technical Research Question:

Optimal Preprocessing Techniques for Grayscale vs. RGB Images: In Section 4.3.8, we discussed the optimal preprocessing techniques for training a CNN on grayscale images as compared to RGB images for emotion recognition and its significance. It was found that converting RGB images to grayscale emphasized shape features over color features, which is crucial for emotion recognition. This conversion also reduced the computational load, as grayscale images have only one channel compared to RGB's three channels.

5.3.4 Implementation and Impact Question:

Practical Implications of Emotion Recognition System in VR Software: Our conclusion (Section 6.1) highlights the potential of integrating emotion recognition systems in VR educational software. Such integration can lead to a more interactive, personalized, and emotionally responsive learning environment. It might alter the landscape of digital learning by enabling real-time assessment of student engagement, thus bridging the gap between digital and traditional educational experiences.

Chapter 6: Conclusion

This chapter summarized the contributions of the thesis and discusses how advanced ML techniques have been integrated into the ever-changing world of VR in education. It provides an overview of the research findings; highlights the approaches that have been used to overcome the challenges of digital learning environments. Additionally, it acknowledges the limitations of this study, providing a transparent overview of the constraints that were encountered. Finally, it suggests potential directions for future research.

6.1 Summary

In concluding this thesis, we reflect on the journey through the world of VR in educational settings. Our focus has been on understanding student engagement and how it intersects with ML techniques. We believe that in today's age where traditional methods of assessing student engagement face challenges, there is a profound need for innovative solutions. We have explored the capabilities of CNNs, particularly a customized training approach model based on the ResNet50 architecture, to train, predict and analyze facial expressions that indicate students' emotional states in VR classrooms. Through our comprehensive experiments, where we compared various neural network architectures and examined the performance of our custom approach CNN model using both full face and partially occluded datasets, we made significant discoveries. The customized approach ResNet50 model proved to be a choice, showcasing commendable accuracy and adaptability across various datasets and scenarios involving facial expression recognition. It even successfully distinguished between expressions of boredom and neutrality.

The research presented here serves as a foundation for future exploration and advancements

in real-time emotion recognition within VR-based educational environments. We have demonstrated that despite the occlusion challenges posed by VR headsets, it is possible to develop a model that accurately captures the lower half of the face. Our customized approach model provides insights for educators regarding student engagement.

In conclusion, it is evident that integrating emotion recognition models into VR platforms has the potential to revolutionize education. By enabling educators to assess student engagement effectively, these technologies can potentially bridge the gap between digital and traditional educational experiences. They can potentially create a more interactive, personalized and emotionally responsive learning environment that could redefine teaching and learning. The research presented in this thesis contributes to the broader discussion on digital education and technology-enhanced learning. It stands as a testament to the potential of ML in creating tools that are both innovative and impactful in tackling the challenges faced by education. With the increasing integration of VR in educational settings, the insights presented in this work will hopefully inspire and guide future research and development. This will ensure that future virtual classrooms are equipped with the necessary tools to promote an engaging, inclusive, and effective learning environment.

6.2 Limitations

There are few limitations we noticed during and after the findings of this study. One key limitation is the lack of data diversity. Despite attempts to include a range of emotional expressions and demographic variations, the datasets used, such as AffectNet and the custom dataset, may not fully capture the extensive spectrum of human emotions and expressions that occur in various educational settings. This limited data diversity could potentially impact the

model's ability to accurately recognize and interpret student emotions universally, especially in ethnically and culturally diverse classrooms.

Another limitation arises from the specificity of the VR environments utilized in this research. While studying simulated VR classroom settings helps facilitate controlled experiments, it might not completely replicate all the complexities and subtleties found in real world VR educational experiences. Real VR classrooms can present unforeseen challenges like fluctuating lighting conditions and variations in VR hardware and software used, factors that could influence the model's performance and applicability in educational settings.

6.3 Future Work

While the current research predominantly utilizes image datasets for training the emotion recognition model, future work can explore the incorporation of video datasets. Video data, encompassing a sequence of frames, offers a dynamic and temporal perspective of facial expressions, which is more representative of real-world interactions in VR classrooms. Training the model on video datasets allows for the capture of the subtleties and progression of emotions over time. Unlike static images, videos can provide context to the transitional phases between different emotional states, offering a deeper understanding of how emotions evolve in response to educational content or interaction within the VR environment.

Recurrent Neural Networks (RNNs), known for their efficacy in handling sequential data, present a promising avenue for future research. RNNs can analyze data where the sequence is important, making them ideal for interpreting facial expressions in video frames, where each frame is a part of a temporal sequence. Therefore, expanding the research to include video datasets and exploring the integration of RNNs offers a promising direction for future work.

Another critical avenue for future research involves expanding data collection efforts. This endeavor would involve not only using the data augmentation technique we described in our methodology but also ensuring a richer diversity in the dataset. Future studies would benefit from capturing a broader array of emotional expressions, possibly including more nuanced or complex emotions that are particularly relevant in educational settings. Additionally, there is a need to include participants from various age groups, cultural backgrounds, and ethnicities to create a more diverse dataset. An expanded dataset would be invaluable in training more robust and inclusive models capable of accurately interpreting and responding to a wider variety of student emotional states within different cultural contexts. Furthermore, incorporating real-time data from actual VR classrooms would provide invaluable insights, enabling training and testing of models in environments that more closely resemble real-world usage. This approach would not only improve the generalizability of the future research but also ensures that the models are attuned to and adaptable to the differences and variations present in live classroom interactions.

In parallel with the expansion of data collection, it becomes crucial to validate the developed models across different platforms. Given the diverse range of VR technologies in use today, it is essential to test and validate emotion recognition models on various VR platforms and headset models. This cross-platform validation ensures that the models are effective not only in theoretical validation but also adaptable and reliable across different hardware and software configurations when tested in actual VR setup. Such validation would involve rigorous testing of these models in various VR environments, encompassing a range of educational applications, headset designs, and interactive modalities. By doing so, future research would address a challenge in the field of VR-based education, ensuring that technological advancements are

beneficial to users across different platforms. This step is essential in moving towards more universally applicable VR educational technology, where the effectiveness of teaching and learning tools are not limited by specific hardware or software requirements.



Figure 6-1: An advanced VIVE VR headset equipped with a camera module, capturing the lower facial expressions of the user. Source: Vive.com

In conclusion, the areas identified for expanding data collection and conducting cross-platform validation present significant opportunities for further research. Addressing these aspects, future studies can make contributions towards developing more accurate, inclusive, and universally applicable emotion recognition models for VR-based educational settings.

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